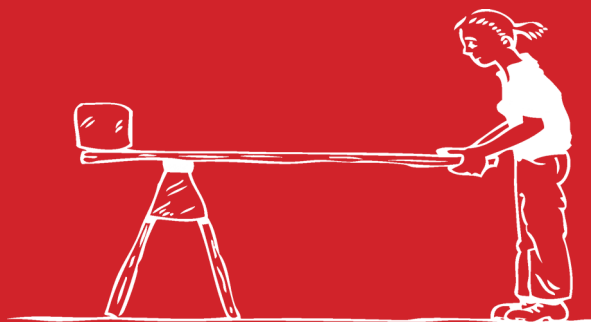


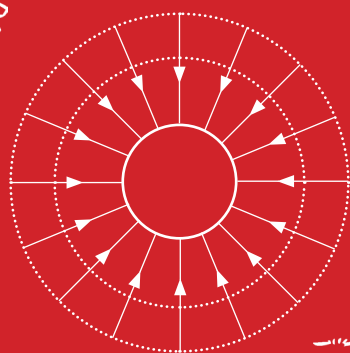
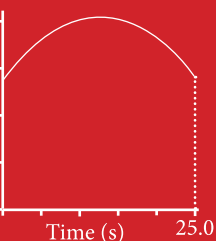
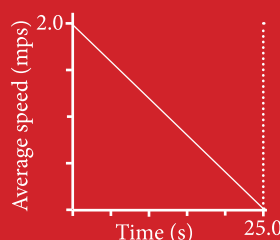
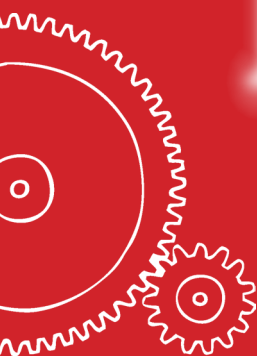
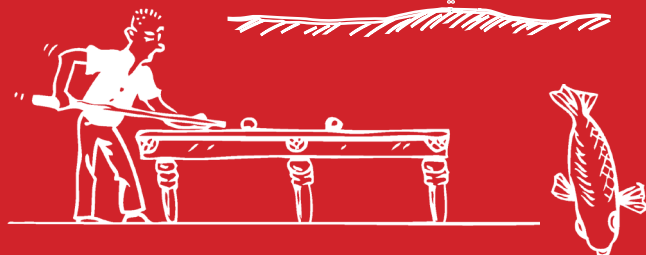
MULTIPLE CHOICE TEST TOPICS



1005 of ready-to-use topic questions!



Physics



Grahame Dobinson

**M U L T I P L E
C H O I C E
T E S T T O P I C S**

1005 of ready-to-use topic questions!

Physics

Grahame Dobinson

This book was commissioned by Jennifer Gough, production was by Piper Print Media, and David Dickson supplied the illustrations.

Grahame Dobinson is a science teacher with many years of experience in teaching all branches of science. He has a genuine love of this subject area, and is an established and successful author who has a long standing association with Blake Education.

The publisher would like to thank Peter Soucek for his assistance with the preparation of this series.

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INTRODUCTION

This series of photocopiable test topic books has been written to assist teachers to set tests and exams for students in years 9–10. The pages and questions can also be used:

- ✦ to prepare students for this specific type of assessment in more senior year levels
- ✦ as a diagnostic tool to check students' understanding of a just completed topic
- ✦ for homework, extension and revision.

The questions in this series ask students to recall or remember a fact, make calculations, interpret data from a table, make sense of a diagram, read off values from a graph, forecast what will happen from a graph and extract information from a written passage.

Each question could be answered in the normal test manner or students could:

- ✦ be asked to justify their response
- ✦ describe why the wrong answers are incorrect
- ✦ provide examples they know that illustrate and support the correct answer.

Teachers should advise students how to record the answers – such as circling the best answer or answering on a separate answer grid. A list of helpful tips for answering multiple choice questions has been provided and was written by teachers well used to setting and marking tests and preparing their students for examinations.

An editable version of all the questions is also available on the CD-ROM that is attached to the inside back cover of this title. All material can be edited only in accordance with the user agreement.

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Before the test or exam

- ☒ Make sure you have planned out how much time you have to answer the multiple choice section of the exam and know exactly how many marks this section is worth.
- ☒ Do as many practice multiple choice questions from past exams as you can.
- ☒ Ensure that you know the meaning of common words found in exams such as *all, many, some, sometimes, always, never, none, not, few*. Also know the definitions of key words such as *describe, analyse* and *evaluate*.

Reading test questions

- ☒ Read through the paper, and select the easiest questions to do first. Do the hardest questions last. This gives you confidence to do your best.
- ☒ Read each question and try to answer it without looking at the choices. Then read through the choices. Doing this allows you to make a clearer and more accurate choice.
- ☒ Be alert for grammatical inconsistencies between the question and the potential answers. A choice is nearly always wrong if the question and the answer don't combine to make a grammatically correct sentence.
- ☒ Don't read anything into or out of the question, or assume it has been written to trick you.
- ☒ Some questions are tricky though. For example, *Which of the following is not ...; All of the following except ...* A response may be true, but not the answer to the question set.
- ☒ The wrong answers are called 'distractors' – there's a reason for this name!

Strategies

- ☒ Eliminate answers you know aren't right and be logical in your approach.
- ☒ With the harder questions, delete the answers that are least likely to be right. Your chances of selecting a correct answer increases with every alternative you eliminate.

- ☒ If there is no penalty for guessing, and you cannot work out the right answer, make learned guesses.
- ☒ Never leave a multiple choice question unanswered.
- ☒ Don't be put off a response just because it seems too obvious or too simple.
- ☒ If two responses appear to be equally correct, eliminate the response that appears to be least related to the question being asked.
- ☒ Use a common sense approach to numerical questions. First think out whether the answer will be in 10s, 100s, 1000s or millions.
- ☒ Unless otherwise instructed, you may write your workings out or reasoning on the test paper and this will not be looked at by the marker.

At the end of the test or exam

- ☒ Leave yourself time at the end of the exam to check your answer sheet to see if you have recorded your answers correctly.
- ☒ Do not change answers unless you know that your recorded answers are incorrect.
- ☒ Do not change your answers just because there are too many or too few of a particular letter on your answer sheet.

After you have been given your marked answer sheet

- ☒ Look carefully at the questions you answered incorrectly. If you answered them wrongly because you did not understand the content, then it is 'back to the books'.
- ☒ If you made an error because your multiple choice test technique was poor, then re-read the tips for answering this kind of question and do more practice.
- ☒ If anxiety meant that you did not perform as well as you wanted, remember that sound planning and a good test technique will help you to feel more confident in any exam. Practice boosts confidence and lessens anxiety.

- 1** The concept of a 'wave motion' is a way of explaining:
- A** how substances can be transferred from one place to another.
 - B** how particles can move through space.
 - C** how energy can be transferred from one place to another.
 - D** all of the above.
- 2** As a wave motion travels through a substance, e.g. sound waves through a brick wall or microwaves through water:
- A** it causes the particles of the substance to vibrate.
 - B** there is always a marked increase in the temperature of the substance.
 - C** it moves the particles a tiny distance from their original positions, permanently.
 - D** it always gathers energy.
- 3** Sound waves will only travel through those substances which are:
- A** solids.
 - B** elastic to some degree.
 - C** either solids or liquids.
 - D** either solids or gases.
- 4** When astronauts on the Moon talk to one another, the sound waves that their voices make can't travel to the other astronaut directly. This is because:
- A** their sounds are stopped by the visors in their helmets.
 - B** their spacesuits are made of a material which stops sounds for safety reasons.
 - C** the pressure inside their spacesuits is too high for sound to be transmitted.
 - D** there is no substance between their two spacesuits to transmit the sound.
- 5** The way that scientists have solved the problem for the astronauts in question 4 is:
- A** to make the visors in the spacesuits from a material which transmits sound.
 - B** to change the sound waves into other waves which can travel from one spacesuit to another.
 - C** to lower the pressure in their spacesuits.
 - D** to make their spacesuits of a different material.
- 6** The larger-than-normal water waves in the oceans which are called 'tsunamis' and were once known as 'tidal waves' are very destructive because they:
- A** are very long.
 - B** are very frequent.
 - C** are very localised.
 - D** possess an enormous amount of energy.
- 7** As the waves created by a speed boat travelling on calm water move and strike a yacht at anchor, the yacht is seen to roll and move up and down, when it was still before. This means that the yacht has gained some extra:
- A** potential energy only.
 - B** kinetic energy.
 - C** large amounts of thermal energy as the waves hit it.
 - D** thermal and potential energy.
- 8** The observation in question 7 could be considered to be evidence supporting the theory that waves:
- A** carry substances from one place to another.
 - B** can transmit both energy and substances.
 - C** can transmit energy.
 - D** can only travel on the surfaces of substances.

9 Unlike sound waves, some waves *can* travel through empty space. These wave motions include:

- ☐ **A** infra-red, light and radio.
- ☐ **B** ultraviolet, light and microwaves.
- ☐ **C** gamma rays, TV and microwaves.
- ☐ **D** all of the above.

10 Some types of waves are known to be very penetrating. Of the following pairs, which are the most penetrating?

- ☐ **A** light and sound waves.
- ☐ **B** radio and infra-red waves.
- ☐ **C** X-rays and gamma rays.
- ☐ **D** TV and ultraviolet.

11 The waves called ‘microwaves’ are used for cooking because they can penetrate into foods, heat them and so cook them. Therefore they are dangerous to humans as a person’s hand would be ‘cooked’ if it were placed in a microwave oven when it was still on. And yet microwaves are used in communication and hence are all around us, constantly bombarding our bodies without harming them. The best explanation for why our bodies are not ‘cooked’ is probably that:

- ☐ **A** the ones used in cooking are much more powerful than the ones used in communication.
- ☐ **B** the ‘cooking’ microwaves are a very different type to the ones used in communication.
- ☐ **C** the ways the two types of waves are transmitted are very different.
- ☐ **D** the substance through which each type passes is very different.

12 Another type of wave motion is called ‘ultrasound’. These waves are so named because they are similar to sound waves, but we can’t hear them. Ultrasound waves are very useful; for example, they are used by medical people to monitor the development of an unborn child as it grows in a pregnant woman’s uterus, and to break up and remove plaque on teeth by dentists. These different uses are best explained by assuming that:

- ☐ **A** ultrasound waves are always harmless to people.
- ☐ **B** the intensity of the ultrasound waves used in medicine is very low and causes no damage.
- ☐ **C** the ultrasound waves used by dentists are a totally different type to those used to monitor an unborn child.
- ☐ **D** all ultrasound waves are harmless to developing embryos, but can break up plaque.

13 Ultrasonic waves in the form of ‘sonar’ devices are used by geologists and fishers to survey the ocean floor and to find schools of fish. A ship sends a signal downwards and detectors on the ship sense when the signal is received at the ship. This could only happen if:

- ☐ **A** the sonar waves can penetrate water and can be reflected.
- ☐ **B** the sonar waves have the same intensity as those used in medicine.
- ☐ **C** the sonar waves are only reflected by rocks and fish.
- ☐ **D** sonar waves are absorbed by water.

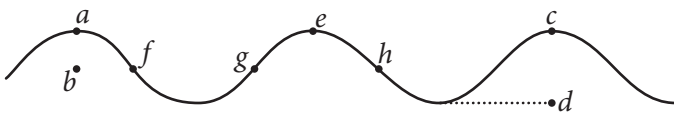
14 It is recognised that there are two basic types of wave motion, *transverse* and *longitudinal*. These two types of waves:

- ☐ A are different because of the ways that they are transmitted through substances.
- ☐ B travel at the same speed through all substances.
- ☐ C are basically the same, but one type has greater wavelengths than the other.
- ☐ D can travel through all types of substances.

15 Examples of the two types of waves in question 14 would be:

- ☐ A ultrasound, X-rays and radio waves.
- ☐ B sonar, sound waves and ultrasound.
- ☐ C light waves, sound waves and microwaves.
- ☐ D A and C above.

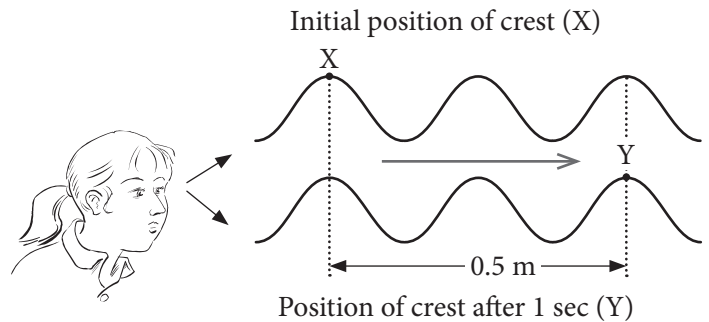
16 Waves produced on water which was once calm can be seen to have *crests* and *troughs*, as shown in the diagram. The disturbed water now transfers energy.



There are terms which are used to describe the wave motion and the way it moves. Two such terms are *amplitude* and *wavelength*. These two terms are correctly indicated by the distances, respectively, between the following points in the above diagram.

- ☐ A a and b, and g and f
- ☐ B a and b, and f and h
- ☐ C c and d, and c and e
- ☐ D c and d, and g and h

Questions 17–20 refer to the following information. An observer in a boat watches waves passing by. She follows the movement of a particular crest and notices that it appears to move to the right by approximately 0.5 metres every second.



17 What she is really observing is:

- ☐ A the wavelength of the wave motion.
- ☐ B the velocity of the wave motion.
- ☐ C the amplitude of the wave motion.
- ☐ D both the amplitude and the wave length of the motion.

18 The observer in question 17 also notices that two waves pass by her in each 1.0 second period. Another way of saying this is that the wave motion:

- ☐ A has a *frequency* of 4.0 waves per second (Hertz).
- ☐ B has a *frequency* of 2.0 Hertz.
- ☐ C has a velocity of 2.0 metres per second (ms^{-1}).
- ☐ D has a velocity of 4.0 metres per second.

19 The wavelength of the motion is:

- ☐ A 0.5 metres.
- ☐ B 0.5 metres per second.
- ☐ C 0.25 metres.
- ☐ D 0.25 metres per second.

20 As the waves pass and the water becomes calm, she estimates that the height from the calm water to a crest would have been 0.1 metre. This measurement represents:

- ☐ A half of the amplitude of the wave motion.
- ☐ B twice the amplitude of the wave motion.
- ☐ C the amplitude of the wave motion.
- ☐ D one quarter of the amplitude of the wave motion, since there were two waves every second.

21 More energy would be transported by a wave motion if it were to have:

- ☐ A a bigger amplitude and frequency.
- ☐ B a smaller velocity.
- ☐ C a larger wavelength.
- ☐ D a smaller amplitude and frequency.

Questions 22–25 refer to the following information. The table shows the results of measurements of the speed of sound waves in different substances, and the densities of those substances. The values in the table are approximate only.

Substance	Air	Water	Iron	Granite rock
Density	0.0013	1.00	7.80	2.6
Speed of sound	330	1410	5130	6000

22 For the speeds in the table to be compared and valid conclusions made:

- ☐ A the units of measurement would have to be the same, where appropriate.
- ☐ B the samples would all have to be taken from the same place.
- ☐ C the speeds would have to be measured under the same conditions.
- ☐ D both A and C would have to be true.

23 Based on the information in the table, which of the following statements is *not* true?

- ☐ A Sound travels faster in water than in air.
- ☐ B The speed of sound through a substance increases with the density of the substance.
- ☐ C Sound appears to travel faster in solids than in liquids or gases.
- ☐ D The speed of sound in granite is greater than in the other substances.

24 A formula linking the speed of sound to the density only of the substance it passes through says that: *The speed of sound is inversely proportional to the square root of its density.* This means that the speed of sound, when considering density alone, would:

- ☐ A increase with the density of the substance.
- ☐ B always be greatest in solids.
- ☐ C decrease with increasing density of the substance.
- ☐ D be hardly affected at all by density.

25 Considering the table and the questions following it suggests that, when trying to assess the speed of sound in different substances:

- ☐ A the data in the table has errors in it.
- ☐ B there are factors to consider other than just density.
- ☐ C temperature is the key factor controlling the speed of sound in a substance.
- ☐ D the sound waves change markedly as they travel through different substances.

26 The reason for classifying wave motions into the types *longitudinal* and *transverse* is that:

- A** in longitudinal waves, the particles of the medium vibrate in the same directional sense as the direction the wave travels; whereas in transverse waves, the particles vibrate at right angles to the direction the waves travel.
- B** in longitudinal waves, the particles of the medium vibrate at right angles to the direction the waves travel; whereas in transverse waves, the particles vibrate in the same directional sense as the direction the waves travel.
- C** longitudinal waves tend to travel lengthways through a substance and transverse waves travel across the substance.
- D** longitudinal waves tend to travel faster through substances than transverse waves.

27 For a wave motion in general, the speed (velocity) of the wave is given by the formula:
velocity = frequency $\times$ wavelength.
 This means that for a particular substance, other things being constant:

- A** wavelength is the main factor in calculating speed.
- B** frequency is the main factor in calculating speed.
- C** the amplitude of the wave also determines its speed.
- D** frequency and wavelength, but not amplitude, determine speed.

28 The standard units used for the variables in the formula $v = f\lambda$ are, respectively:

- A** metres, hertz (cycles per second) and metres per second.
- B** hertz, metres and metres per second.
- C** metres per second, hertz and metres.
- D** metres, metres per second and hertz.

29 The relationship $v = f\lambda$ indicates that, for a wave motion, the velocity or speed:

- A** increases only if f and λ both increase.
- B** increases if either f or λ increases, provided that the other one does not decrease.
- C** is constant if one increases and the other decreases.
- D** only increases if f decreases.

30 Which statement is true about the formula $v = f\lambda$, as it applies to a particular wave motion?

- A** v could increase even if f decreased, provided that λ increased.
- B** v could not increase if f decreased, no matter what.
- C** v would increase even if f decreased, providing that λ increased.
- D** v would increase if f were halved and λ were doubled.

31 If the speed of sound in air is 330 metres per second and a wave motion of frequency 512 hertz is passed through the air, then the wavelength of the wave motion would be:

- A** 512×330 metres.
- B** 0.64 metres per second.
- C** 0.64 metres.
- D** $512 / 330$ metres.

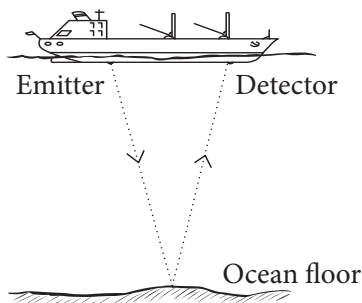
32 If a particular ultrasound has a frequency of 25 000 hertz and travels at the velocity or speed of 330 metres per second, then the wavelength of the ultrasound would be closest to:

- ☐ A 0.1 metres.
- ☐ B 0.1 metres per second.
- ☐ C approximately 76 metres.
- ☐ D 0.01 metres.

33 A large wave motion on water is estimated to have a wavelength of 4.0 metres, an amplitude of 3.0 metres and a frequency of 0.1 hertz. This means that the velocity of the wave motion would be:

- ☐ A 0.4 metres per second.
- ☐ B 12.0 metres per second.
- ☐ C 0.3 metres per second.
- ☐ D 1.3 metres per second.

34 A particular 'echo sounder' on a ship uses ultrasound waves of frequency 30 000 hertz and wavelength 0.048 m. The waves leave the hull of the ship, head to the ocean floor and are reflected. They are picked up by a detector on the ship, which allows the depth of the ocean floor to be calculated.



The additional information which would have to be supplied to calculate the depth of the water is:

- ☐ A the amplitude of the waves used.
- ☐ B the velocity of the waves in seawater.
- ☐ C the time for the waves to go to the bottom and back to the ship.
- ☐ D the temperature of the water.

35 The echo sounder operator in question 34 calculates the depth to the ocean floor to be 150 metres. This means that the time of travel of the ultrasound waves must have been closest to:

- ☐ A 0.4 s.
- ☐ B 0.2 s.
- ☐ C 0.8 s.
- ☐ D 0.1 s.

36 The 'electromagnetic spectrum' could best be described as a broad range of wave motions which:

- ☐ A travel at the same speed and are all of the same type.
- ☐ B travel at the same speed and have the same penetrating ability.
- ☐ C show a variety of ways by which they move through substances.
- ☐ D have the ability to travel through space but not through liquids.

37 The fact that any type of electromagnetic radiation is a wave motion means that any particular example would be able to be:

- ☐ A reflected and refracted, but not diffracted.
- ☐ B reflected, refracted and diffracted.
- ☐ C refracted and diffracted, but not reflected.
- ☐ D diffracted and reflected but not refracted.

38 The velocity of electromagnetic waves in space is 3×10^8 metres per second. The velocity of the same waves in air would be:

- A** a lot less because of the density of the air.
- B** more because the electromagnetic waves have a substance to help them travel.
- C** a little less because of the density of the air.
- D** a little more because air is a very elastic substance.

39 The different electromagnetic waves are given different names to identify their respective places in the electromagnetic spectrum. These names are chosen based on:

- A** the different speeds at which they travel in different substances.
- B** the wavelengths of the particular electromagnetic radiation band.
- C** the amplitudes of the particular electromagnetic radiation.
- D** the source of the particular electromagnetic radiation.

40 Which list of wave motions below contains only two types that are electromagnetic waves?

- A** radio, sonar, light and ultrasound
- B** radio, ultra-violet, ultrasound and microwaves
- C** light, microwaves, TV waves and X-rays
- D** sound, light, infra-red and gamma rays

41 The speed of sound in air is approximately 330 mps while the speed of light is approximately 3×10^8 mps. Early attempts to measure the speed of light showed that it was quite difficult to do, much more difficult than measuring the speed of sound. This was probably because:

- A** sound is easily detected while light is not.
- B** it was difficult to obtain a source of light bright enough for measurements to be made.
- C** the speed of light is enormous compared to the speed of sound.
- D** sound waves are more easily reflected than light waves.

42 The electromagnetic spectrum covers a very wide range of frequencies and wavelengths, therefore the measurement of wavelengths requires the use of both extremely tiny units and extremely large ones. For this reason, appropriate prefixes are used with the unit 'metre'. Three prefixes used in this way to describe the very large and the very small wavelengths are:

- A** *nano*, meaning 10^{-9} or $1/10^9$, *milli*, meaning 10^{-3} or $1/10^3$ and *kilo*, meaning 10^2 .
- B** *nano*, meaning 10^{-9} or $1/10^9$, *micro*, meaning 10^{-6} or $1/10^6$ and *kilo*, meaning 10^3 .
- C** *micro*, meaning 10^{-6} or $1/10^6$, *milli*, meaning 10^{-4} or $1/10^4$ and *kilo*, meaning 10^3 .
- D** *nano*, meaning 10^{-9} or $1/10^9$, *kilo*, meaning 10^3 and *mega*, meaning 10^6 or $1/10^6$.

- 4|3** The wavelengths of some typical electromagnetic waves are listed in the following table, measured in metres. Which of the prefixes used in question 42 would be most appropriate, respectively, for the three examples in the table?

Type of wave	Long radio wave	Microwave	X-rays
Example of wavelength	2000	0.03	2×10^{-10}

- A** kilo, micro, nano
B kilo, milli, nano
C kilo, micro, milli
D milli, micro, micro
- 4|4** If microwaves and ultraviolet waves left a source one second apart, with the microwaves leaving first, and travelled in the same medium, they would arrive at detectors:
- A** one second apart because their velocities are the same.
B at the same time because the ultraviolet waves would catch up to the microwaves.
C further apart because of the greater wavelength of the microwaves.
D at the same time, but only if they were travelling in air.
- 4|5** The energy carried by electromagnetic waves is directly proportional to their frequencies. This means that the correct ranking of the following waves from those carrying the most energy to those carrying the least, other factors being constant, is:
- A** X-rays, microwaves, ultraviolet waves.
B microwaves, radio waves, infra-red.
C ultraviolet, radio waves and infra-red.
D gamma rays, X-rays, radio waves.

- 4|6** The Earth's atmosphere partially blocks the Sun's rays, affecting the various wavelengths to different degrees. For this reason, the most accurate measurements of the Sun's electromagnetic emissions are done:

- A** from aeroplanes that fly at high altitudes.
B from satellites that orbit the Earth above the limits of the atmosphere.
C from very high mountains as the radiation detectors need to be on the Earth.
D at high altitudes but within the Earth's atmosphere to give recording instruments some protection.

- 4|7** The science of 'radio astronomy' is so named because the astronomers use detectors of incoming radio waves from distant stars and other bodies rather than light waves. It is a relatively new science compared to 'optical astronomy'. This would suggest that:

- A** the stars which emit radio waves are fairly new ones.
B all stars give out radio waves, but not all of the time, so it is hard to detect them.
C the detection of radio waves gives astronomers a new tool to work with in studying the heavens.
D the emission of radio waves by stars, etc., has only started in recent times.

- 4|8** Three 'detectors' which each detect a particular band of electromagnetic radiation are:

- A** our eyes, our ears and our skin.
B our eyes, radio telescopes and TV antennae.
C our ears, infra-red remote controls and optical telescopes.
D X-ray machines, police radar machines and sonar detectors.

49 Two types of electromagnetic waves emitted by the Sun, one which is fairly well blocked by the Earth's atmosphere and one which penetrates the atmosphere very well are, respectively:

- A** light waves and gamma rays.
- B** radio waves and ultraviolet rays.
- C** light waves and ultraviolet rays.
- D** ultraviolet radiation and light waves.

50 The 'greenhouse effect', as observed in a glasshouse, is a result of the fact that:

- A** all electromagnetic waves, after passing through glass, cannot be reflected back.
- B** infra-red rays pass through glass, but the same waves cannot be reflected back out.
- C** gamma and X-rays pass through glass and are absorbed so that they cannot be re-radiated.
- D** some infra-red rays pass through glass, are absorbed and re-emitted with different wavelengths that cannot penetrate glass.

51 Microwaves are not able to pass totally through water. Instead, they are absorbed by water and converted to some other form of energy. This explains why microwaves:

- A** can heat foods which usually contain a lot of water and so cook them.
- B** lose power when used for communication in space.
- C** are used for communication purposes, but only where the moisture level is zero.
- D** are not used for remote controls as they would never reach their targets.

52 The remote controls which we use to operate TVs and DVD players usually use infra-red waves. The fact that we can use these remotes in the home means that:

- A** other wave types cannot penetrate the air between the remote and the detector.
- B** the infra-red rays used can penetrate the normal air in a household but are not powerful enough to be harmful.
- C** the penetrating powers of other electromagnetic waves are not good enough to be used.
- D** infra-red waves can never be of an intensity that can cause problems.

53 Electromagnetic waves cannot travel through metals. This explains why:

- A** foods are not to be wrapped in 'foil' prior to cooking in a microwave.
- B** the radio aerial of a car is usually outside the vehicle and is connected to the radio by a wire.
- C** metals are used as 'blocking devices' to stop unwanted electromagnetic radiations.
- D** all of the above.

54 X-rays used in medicine can help to detect abnormalities in our bodies, particularly in relation to the bones. This is most likely due to the fact that X-rays:

- A** pass through all body tissues with equal ability.
- B** are totally harmless, regardless of their power or intensity.
- C** pass through different tissues to different degrees, hence enabling abnormalities to be identified.
- D** are the only types of electromagnetic waves which can pass through all human tissues.

1 When average people speak of ‘electricity’ in general terms, they are most likely referring to:

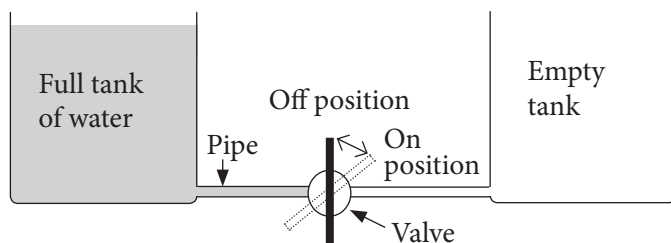
- A** the electric charge which can exist on many objects.
- B** the passage of electrical charge through a conducting material.
- C** a form of energy that can only be produced by generators.
- D** energy which can only travel through wires and always causes heat to form.

2 An electric circuit with a current flowing is often compared to a flow of water. This is because:

- A** electricity, like water, is a flow of a substance through a conductor.
- B** most of the factors involving the flow of electricity are similar, in principle, to those involving a flow of water.
- C** electricity needs only a conductor to cause a flow of current.
- D** when electricity flows there is no opposition, just like water.

Questions 3–8 refer to the following information.

Consider two water tanks, one full and one empty, connected by a pipe with a ‘valve’ in it which can turn water flow on or turn it off. The following diagram summarises the situation.



3 Suppose that the ‘valve’ is now turned to the ‘on’ position. It would be true to say that, in this diagram:

- A** the full tank is similar to the ‘destination’ in electricity.
- B** the ‘valve’ is similar to a switch in electricity, while the empty tank is like a ‘source’ of electricity.
- C** the full tank is like a ‘source’ of electricity which supplies pressure even when the ‘valve’ is turned off.
- D** the empty tank is like a ‘source’ of electricity, but only while the ‘valve’ is turned off.

4 The ‘pipe’ and the ‘valve’ in the diagram play roles which are similar to the roles played in electricity by, respectively:

- A** the switch and a conducting wire.
- B** the current flow and the switch.
- C** the conducting wire and the switch.
- D** the conducting wire and a source of electricity.

5 When the valve is turned to the ‘on’ position:

- A** water will flow from left to right until the once-empty tank is full.
- B** water will flow at a constant rate until the full tank is empty.
- C** water will flow to the empty tank until it is full and then flow back.
- D** none of the above will happen.

- 6** A true statement about the flow of water when the valve is turned on is:
- A** the pressure on the water increases as the water flows.
 - B** the flow is caused by the fact that the water in the tank on the left has potential energy.
 - C** the same pressure is maintained on the water until the tank is empty.
 - D** the flow will continue, even when the two tanks hold equal amounts of water.

- 7** Which statement is true about the situation when the valve is turned on?
- A** The amount of water flowing will steadily decrease until it stops.
 - B** The flow of water will stop when the level of water in both tanks is the same.
 - C** The flow of water will stop when the level of water in the tank on the right is a little higher than that in the tank on the left.
 - D** Statements A and B above are both true.

- 8** If a pump and a second pipe were now fitted to return water to the tank on the left as fast as it was moving through the open valve to the tank on the right, then:
- A** the pressure causing water flow would be a constant.
 - B** the same amounts of water would move through the valve in equal intervals of time.
 - C** both A and B would be true.
 - D** the amount of potential energy in the water in the tank on the left would gradually increase.

- 9** When a water tap is turned off, the pressure trying to push water out of the tap is still there, even though we don't see its effects. When electricity is turned off using a switch:
- A** a small, harmless amount of electricity always still flows.
 - B** the electrical pressure quickly drops to zero.
 - C** the effect on the circuit is to eliminate the resistance and the electrical pressure.
 - D** the effect is similar to turning off the water tap and the electrical pressure is still there.

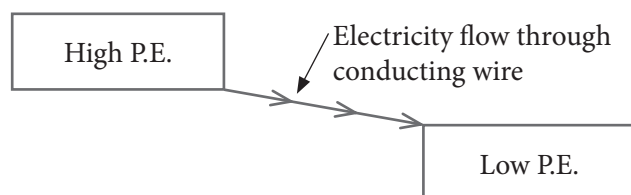
- 10** The voltage (potential difference) present in an electrical circuit is somewhat like:
- A** measuring the amount of water which actually passes out of a tap when it is turned on.
 - B** the pressure in water which forces a flow when a tap is turned on.
 - C** the restriction which a hose causes to the flow of water.
 - D** the speed with which water can leave a tap when it is turned on.

- 11** The potential difference applied to a circuit is:
- A** also called the 'electromotive force' or 'emf'.
 - B** a measure of the force trying to push electricity through a circuit.
 - C** usually measured in units called 'volts'.
 - D** all of the above.

12 The 'resistance' present in a circuit is really a measure of:

- A** how much the voltage opposes the flow of current.
- B** both the potential difference and the time taken for the electricity to flow around a circuit.
- C** the opposition to the flow of current in the circuit.
- D** the effect of the current flowing in a circuit, since the resistance varies greatly as the current varies.

13 When electricity flows, it flows from a region of high potential energy to a region of low potential energy, hence the term 'potential difference'. That is:



As this happens it would mean that:

- A** the potential energy of the source would lessen and the energy stored in the 'destination' would increase.
- B** the electrical flow would soon be reversed.
- C** regardless of the amount of electrical flow, the amount of potential energy in both the source and the destination would remain the same.
- D** the potential energy of the source gradually increases.

14 In the situation described in question 13, if the electricity flow was to be a continuous one, then:

- A** a potential difference between source and destination would have to be maintained.
- B** the resistance of the circuit would have to be almost negligible.
- C** the conducting wire would have to be shortened considerably.
- D** the conducting wire would have to be replaced by one which does not conduct very well.

15 An electric current passing through a conducting solid:

- A** is really a flow of atoms of the conducting material.
- B** is a flow of ions from the source of electricity through the solid conductor.
- C** has been shown to be a flow of electrons through the conducting solid.
- D** is known to be a flow of electrons which comes only from the source of electricity and does not involve the conducting solid.

16 The only elements which can conduct electricity to such a degree that they can truly be called 'conductors' are:

- A** metals and the non-metal carbon.
- B** metals and the three non-metals carbon, sulphur and phosphorus.
- C** a few of the 'heavy metals' such as gold, silver and copper.
- D** some selected metals which are the ones used in residential and commercial circuitry.

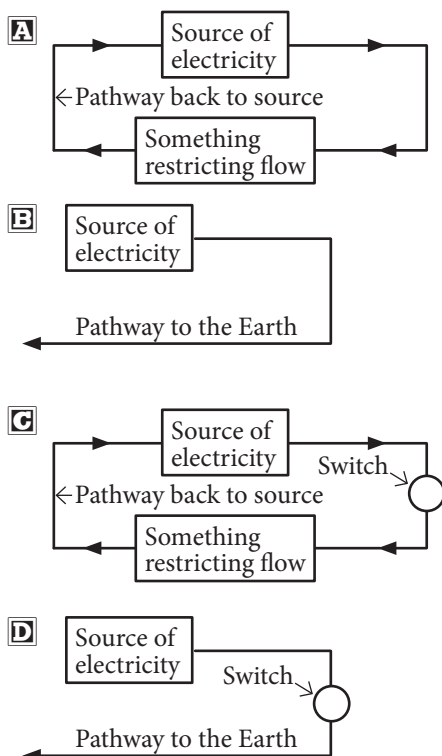
17 Electric currents, in general, are:

- A** all really the same in principle, no matter how large or small they are.
- B** basically different, depending on the nature of the material through which they pass.
- C** hard to detect when they are very small.
- D** unable to be tolerated by the human body, even in the most minute amounts.

18 By the term 'an electric circuit', we mean:

- A** a source of electricity such as the power supply to our homes.
- B** a source of electricity which could be from anywhere.
- C** a source of electricity and a pathway so that the electricity can flow back to its source.
- D** a pathway for electricity which offers absolutely no opposition to the flow of electricity.

19 Which diagram shows the best representation of the essentials of an electric circuit?



20 Which combination in the table gives the best matched comparisons between water flow and electrical flow?

Water terms	Electrical terms
P – pressure in water	X – resistance
Q – amount of water that flows	Y – potential difference
R – a 'kink' in a hose which restricts the flow	Z – current

- A** P and Y, Q and X, R and Z
- B** P and Y, R and X, Q and Z
- C** P and X, Q and Y, R and Z
- D** P and Z, Q and Y, R and X

21

Quantity measured	Units used
1 – potential difference	X – amperes
2 – current	Y – volts
3 – resistance	Z – ohms

The combination from the table above that correctly links the electrical quantities we measure with the units used for measuring them is:

- A** 1 and X, 2 and Y, 3 and Z.
- B** 1 and Z, 2 and Y, 3 and X.
- C** 1 and Y, 2 and X, 3 and Z.
- D** 1 and Y, 2 and Z, 3 and X.

22 Common sources of potential difference used in our society to force electricity through circuits include:

- A** generators, solar cells and batteries.
- B** generators, batteries, solar heaters and electric motors.
- C** solar hot water systems, batteries and electric radiators.
- D** batteries, electric motors and electric heating devices.

2|3 It would be true to say that, in an electric circuit:

- ☐ A as the length of wires in the circuit becomes greater, a bigger potential difference is needed to maintain a constant current flow.
- ☐ B there is no loss of energy as the electricity travels around the circuit.
- ☐ C there is a loss of electrons as current flows through the circuit.
- ☐ D there is no need to supply a potential difference once the current starts to flow.

2|4 Batteries are generally rated as AAA, AA, A, B, C, and D, according to their size, even though they may all be of potential difference 1.5 volts. This probably means that:

- ☐ A the larger size D batteries can only supply large amounts of electric currents.
- ☐ B the different size batteries all have the same amount of stored electrical energy and are made in different sizes just to fit into different size appliances.
- ☐ C the larger batteries store more energy than the smaller ones but apply the same 'pressure' to an electrical circuit.
- ☐ D it is always impossible to use a 'D' size battery where a 'C' size is recommended.

2|5 The electrical energy in any cell or 'battery' of cells:

- ☐ A is an amount of separated electrons which are 'pumped' into the cell or battery before it is sealed at the factory.
- ☐ B is produced by chemical reactions within the cell or battery.
- ☐ C can be large or small and is independent of the battery size.
- ☐ D can be contained within the battery for an indefinite period of time.

2|6 When placing AAA batteries into a remote control:

- ☐ A the '+' end of a battery should always go into the '-' end of the remote control.
- ☐ B only the '+' end of the battery needs to make contact with the metal contact in the remote control.
- ☐ C if the '+' and '-' ends are mixed up then the remote control will draw energy into it.
- ☐ D the '+' end of the battery must be connected to the '+' contact in the remote control.

2|7 When a battery is described as being 'flat', it means that:

- ☐ A the potential difference between the '+' and the '-' is zero so no current will flow.
- ☐ B the chemicals in the battery have all been converted into electrical energy.
- ☐ C the '+' and the '-' terminals have both become positive.
- ☐ D the '+' and the '-' terminals have both become negative.

2|8 A mechanical 'generator' of electricity:

- ☐ A basically converts kinetic energy into electrical energy.
- ☐ B uses the same basic principle, whether it is one in a hydro-electric power station or one in a motor vehicle.
- ☐ C needs some form of magnet in it to be productive.
- ☐ D has all of the above features.

30 When we speak of a 'direct current', we mean one which:

- A** passes through wires without any, or only very low, resistance.
- B** travels the shortest possible pathway through a circuit.
- C** travels only one way through a circuit and does not reverse its pathway.
- D** is produced only by generators and not by batteries.

30 The currents we call 'AC' currents, meaning 'Alternating Currents':

- A** are so named because the current flows back and forth in the circuit, usually changing direction at regular intervals.
- B** are relatively useless for heating purposes.
- C** are produced by rechargeable batteries because the chemical reactions can be reversed regularly and quickly.
- D** are not produced by any type of generator, but only by special types of cells.

31 There are many devices which can or must be included in an electric DC circuit. To simplify things, 'symbols' are used to identify the devices. Four such symbols are shown in the table.

Symbol	A	B	C	D
Meaning	conductor	potential difference	switch	resistor

One of these symbols is incorrect. It is the one labelled:

- A** A. **B** B. **C** C. **D** D.

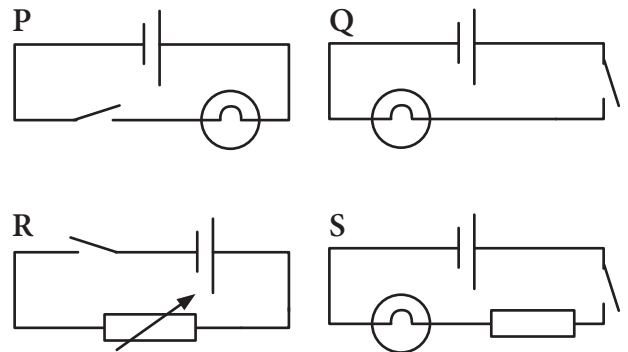
32 Some additional devices which can be present in circuits are shown by the following symbols.

Symbol	I	II or	III	IV
Meaning	variable emf	variable resistor	voltmeter	wires not touching

The incorrect symbol in the table is:

- A** I **B** II **C** III **D** IV

33 Which diagram is the correct representation of a simple circuit involving a source of potential difference, a light bulb, a switch and some copper conducting wires?



- A** P **B** Q **C** R **D** S

34 Which diagram shows a switch in the correct position?

Diagram A

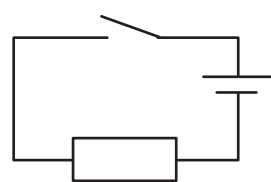


Diagram B

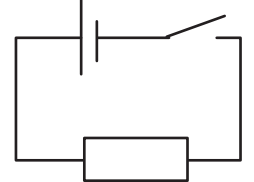


Diagram C

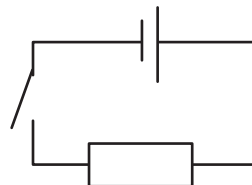
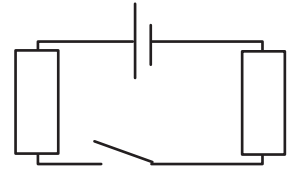


Diagram D



- A** Diagram A **B** Diagram B
- C** Diagram C **D** Diagram D

35 Devices which are really ‘variable resistors’ and are used in the home include:

- ☐ A fluorescent light bulbs, volume controls on radios and dimmer switches on lights.
- ☐ B incandescent light bulbs, temperature controls on electric frypans and volume controls on radios.
- ☐ C volume controls on radios, temperature controls on electric frypans and dimmer switches on lights.
- ☐ D both incandescent and fluorescent light bulbs and volume controls on frypans.

36 In an electric circuit, it is best to always place the switch next to:

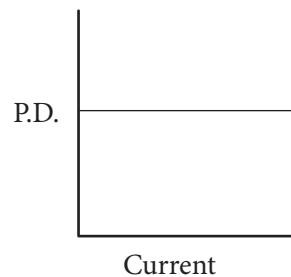
- ☐ A the ‘–’ terminal of a power source as this is where the electrons come from.
- ☐ B the ‘+’ terminal of a power source since this cuts the high potential being applied to the rest of the circuit.
- ☐ C the ‘+’ terminal of a power source since this prevents any ‘back surge’ from the ‘–’ terminal.
- ☐ D any component since it is irrelevant where the switch must be.

37 Ohm’s Law is a relationship in electricity which, in simple terms, shows that:

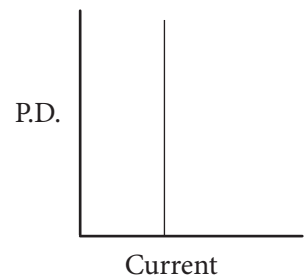
- ☐ A as the potential difference applied to the ends of a conductor is increased, the current passing through the conductor remains a constant.
- ☐ B the resistance of a conductor depends only upon the current passing through it.
- ☐ C the current passing through a conductor depends on its resistance and is independent of the potential difference applied across its ends.
- ☐ D there is a direct relationship between the current flowing through a conductor and the potential difference across its ends.

38 Which graph correctly shows the relationship between the current flowing through a conductor and the potential difference (P.D.) across the ends of the conductor?

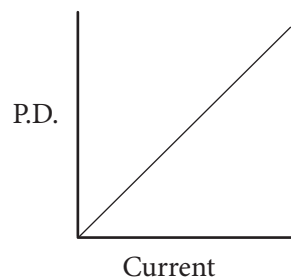
Graph 1



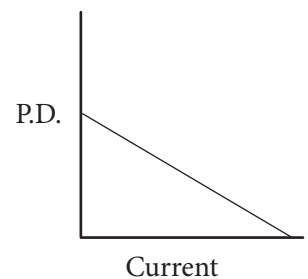
Graph 2



Graph 3



Graph 4



- ☐ A Graph 1
- ☐ B Graph 2
- ☐ C Graph 3
- ☐ D Graph 4

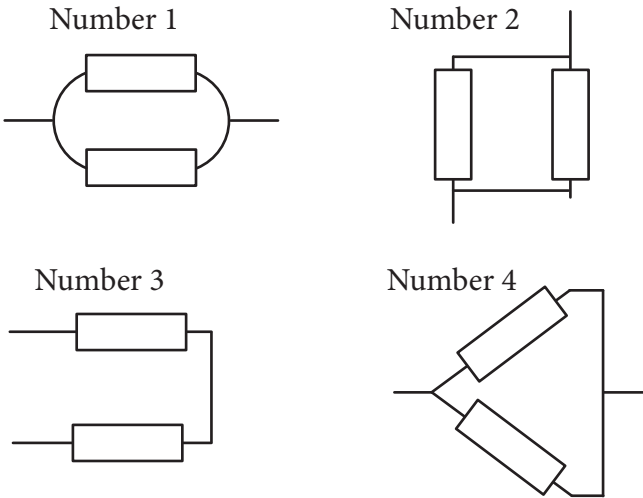
39 Of the graphs shown in question 38, the one which shows that ‘as P.D. is increased, the current flowing is decreased’ is:

- ☐ A Graph 1.
- ☐ B Graph 2.
- ☐ C Graph 3.
- ☐ D Graph 4.

40 When two electrical devices are connected in series, it means that:

- ☐ A they are connected so that there is only one pathway through which the current can pass.
- ☐ B the two devices each have a separate pathway through which electricity can pass.
- ☐ C they are not able to be connected in a ‘parallel’ type of connection.
- ☐ D the P.D. applied to each of them is the same.

- 41** Which diagram shows an example of two resistors connected in series?



- A** Number 1 **B** Number 2
C Number 3 **D** Number 4

- 42** If the ends of the connections in the four situations in question 41 were connected to identical sources of P.D. then the same P.D. would be applied to the pair of resistors in diagram:

- A** number 1 only.
B numbers 2 and 4.
C numbers 1, 2 and 4.
D number 3 only.

- 43** If a P.D. of 12 volts applied to the ends of a piece of wire causes a current of 2 amperes to flow through the wire, then it would be true to say that:

- A** the wire has a resistance of 24 ohms and when a P.D. of 24 volts is applied to the wire, a current of 4 amperes would flow.
B the wire has a resistance of 6 ohms and when a P.D. of 24 volts is applied to the wire, a current of 4 amperes would flow.
C the wire has a resistance of 6 ohms but when a P.D. of 24 volts is applied, its resistance drops markedly.
D the resistance of the wire is 6 volts and when a P.D. of 36 volts is applied to it, then a current of 6 amperes will flow.

- 44** Two resistors, each 10 ohms, are connected (a) in series and afterwards (b) in parallel. A true statement about their effects when combined in the different ways and then placed in a circuit is:

- A** the (a) type connection would cause the total resistance when placed in a circuit to be less than 10 ohms.
B the (b) type connection would make the total resistance of the combination 20 ohms.
C there is no difference between the (a) and (b) type connections on the total resistance when placed in a circuit.
D the (a) type connection would double the 10 ohm resistance in the circuit, while the (b) type would halve it.

- 45** The system of 'parallel' connections is used in motor vehicle lighting, because the method:

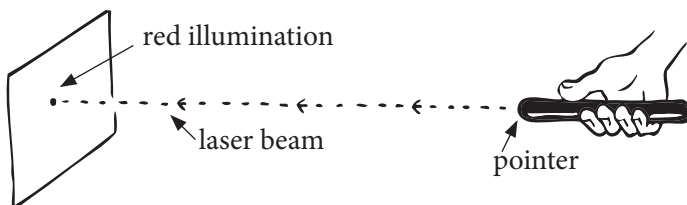
- A** reduces the amount of energy that has to be supplied by the car's battery.
B is much safer to use with a 12 volt battery than the series method.
C is the only method which can be used when supplies are as low as 12 volts.
D allows all other light bulbs in the vehicle to function even if one bulb doesn't work.

- 46** The following table shows some features of ammeters and voltmeters. Which combination of features correctly applies to I – an ammeter, and II – a voltmeter?

Resistance of meter	What the meter measures	How it is connected into a circuit
(a) High	(b) Potential difference	(c) In series
(d) Low	(e) Electric current	(f) In parallel

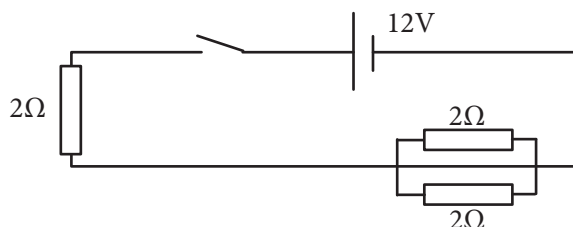
- A** I – (a), (e), (c) and II – (a), (b), (f)
B I – (d), (e), (c) and II – (d), (b), (c)
C I – (d), (e), (c) and II – (a), (b), (f)
D I – (d), (b), (c) and II – (a), (b), (f)

- 4|7** A small, hand-held laser pointer, often used by presenters and lecturers, is powered by two AAA batteries, each of potential difference 1.5 volts, connected in series.



The power output of the laser pointer is listed as 3.0 mW. This means that it is operating:

- A** with a potential difference of 3.0 volts and uses a current of 1×10^{-3} A.
 - B** with a potential difference of 1.5 volts and uses a current of 1×10^{-3} A.
 - C** with a potential difference of 3.0 volts and uses a current of 1 A.
 - D** with a potential difference of 1.5 volts and uses a current of 1 mA.
- 4|8** A circuit is set up with a 12V power supply, a switch and three 2 ohm resistors, as shown in the diagram.



For this circuit, it is true to say that (a) the total resistance, (b) the current flowing through, and (c) the power of the circuit are, respectively:

- A** 6 ohm, 2 amp, 24 watt.
- B** 4 ohm, 3 amp, 12 watt.
- C** 3 volt, 4 amp, 48 watt.
- D** 3 ohm, 4 amp, 48 watt.

- 4|9** The headlights on a motor car are each labelled '12 V, 60 W'. If one happens to 'blow', the other headlight still works. A true statement about this situation is that:

- A** the bulbs are connected in parallel and, when one blows, the total current still used is 10 A.
- B** the bulbs are connected in parallel and, when both bulbs are working, the total current from the battery is 10 A.
- C** the power of the circuit when one bulb is working is 60 W and the bulbs are connected in series.
- D** the two bulbs, when working together, would require a potential difference of only 6 V each since the combined resistance is halved.

- 5|0** When the rechargeable battery pack of a 12 V electric drill is taken apart, it is found to consist of 10 individual smaller cells connected to one another. This probably means that each of the 10 cells:

- A** would be 1.2 V, and is connected in parallel.
- B** is approximately 1.5 V to allow for loss due to heat.
- C** would be 1.2 V and they are connected in series.
- D** is 1.2 V and the method of connection is irrelevant, since they will always produce 12 V.

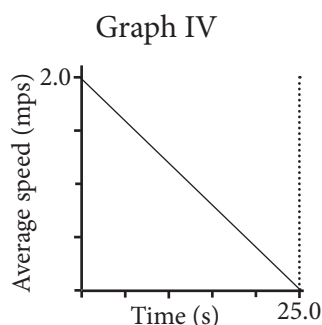
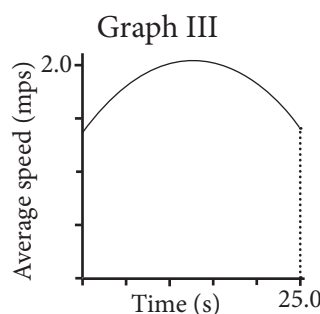
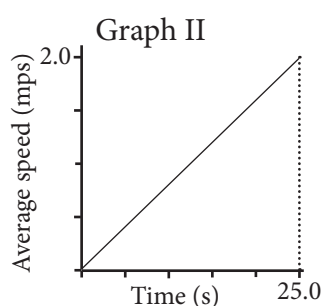
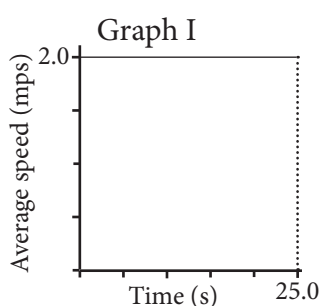
1 The terms *speed* and *velocity* are used to describe objects which are moving. They:

- A** mean exactly the same thing and are measured with the same units.
- B** are similar terms, measured with the same units, but are not exactly the same.
- C** both involve measurements of distance, time and direction.
- D** are used in different situations, with velocity being used only when the values are very large.

2 If a swimmer completes the length of a 50 m Olympic Pool in 25.0 s then it means that the swimmer:

- A** maintained a constant speed of 2.0 mps.
- B** probably swam with a speed greater than 2.0 mps during the first 25 m.
- C** had an average speed of 2.0 mps.
- D** maintained a constant velocity of 2.0 mps for all of the swim.

3 Which graph correctly shows the motion of the swimmer in question 2?

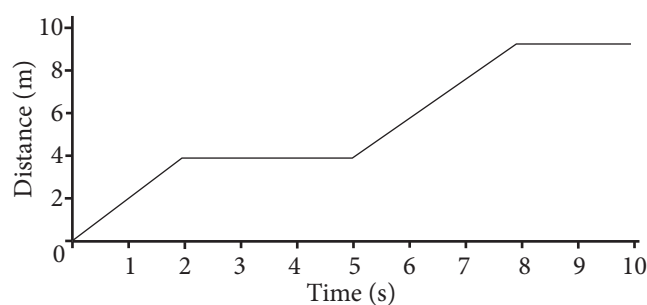


- A** Graph I
- B** Graph II
- C** Graph III
- D** Graph IV

4 One of the graphs in question 3 shows a swimmer's average speed decreasing with time. A second graph shows a swimmer's average speed increasing with time. These two graphs are, respectively:

- A** I and II.
- B** II and III.
- C** I and III.
- D** IV and II.

5 A common graph used to represent the motion of an object is a 'Distance vs Time' graph. In the graph below, an object is tracked from its original position for 10 s, with its position being noted each second.



A correct interpretation of this graph is that the object:

- A** did not cover any distance (i.e. speed = 0) for 5 of the 10 seconds.
- B** moved with a fairly constant speed.
- C** covered a distance of 8 m in 6 s.
- D** was stationary for more time than it was moving.

6 In a graph such as that in question 5, the *slope* or *gradient* of the graph shows the:

- A** distance the object moved in the time interval concerned.
- B** speed of the object at any given time.
- C** possible forward or backward motion of the object.
- D** rate of change of the speed.

7 In the 2007 Australian Motor Cycle Grand Prix at Phillip Island, Victoria, Australia's Casey Stoner did a lap time of 1 minute 31.13 seconds for the 4.4 km circuit. This means that his *average speed* was:

- A** 4.4 / 91.13 mps.
- B** $4.4 \times 60 \times 60 / 91.13$ kmph.
- C** $4.4 \times 10^3 \times 60 \times 60 / 91.13$ kmph.
- D** $4.4 \times 60 / 91.913$ mps.

8 In a handicap event such as the professional 100 m footrace events the 'Stawell Gift' or the 'Botany Gift', the runners are given starts according to their 'abilities'. The slowest runner in a particular event is given a start of 10 m while the fastest, or 'scratch' runner has to run the full 100 m. These two runners 'dead heat' in the race, in a time of 10.0 s. Their average speeds in metres per second are correctly shown in which alternative in the following table?

Runner	I	II	III	IV
Scratch	10.0	9.0	10.0	10.1
Slowest	8.9	10.0	9.0	9.0

- A** I **B** II **C** III **D** IV

9 The average speed of a snail is estimated to be 0.002 mps. At this rate, in 1 minute a snail would cover a distance of:

- A** 0.012 m. **B** 0.010 m.
- C** 1.2 cm. **D** 0.12 m.

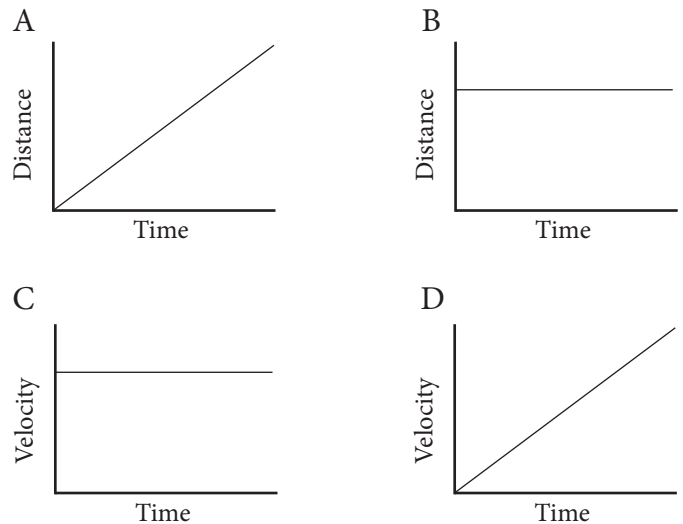
10 A typical cruising speed for a jet aircraft is 150 mps. This would mean that the time taken for the jet to travel 1000 km is:

- A** $(1000 \times 1000 / 150)$ seconds.
- B** $(1000 / 150)$ seconds.
- C** $(1000 \times 60 / 150)$ seconds.
- D** $(1000 \times 1000 / 60 \times 60)$ seconds.

11 When a moving object is said to 'accelerate' it means that:

- A** its velocity increases and then falls back below its original value.
- B** its velocity changes.
- C** its speed increases but only by very small amounts.
- D** the only change is in the direction of the object's motion.

12 Which graph shows that an object is accelerating as time passes?

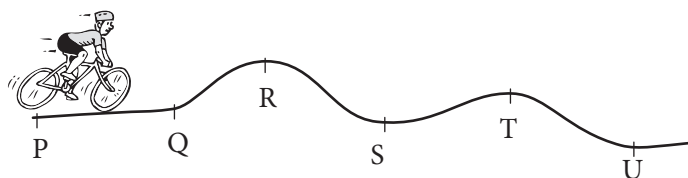


- A** A **B** B **C** C **D** D

13 A car driver watches her 'speedo' as she starts to drive and sees that the speed indicated at different times in km per hour is: 10, 30, 50, 60, 60, 60. This means that her car:

- A** was accelerating all of the time.
- B** moved with constant speed all of the time.
- C** accelerated for a while and then moved with a constant speed.
- D** had a constant speed for a while and then accelerated.

- 14** A cyclist rides a hilly course between points P and U, as shown in the diagram.

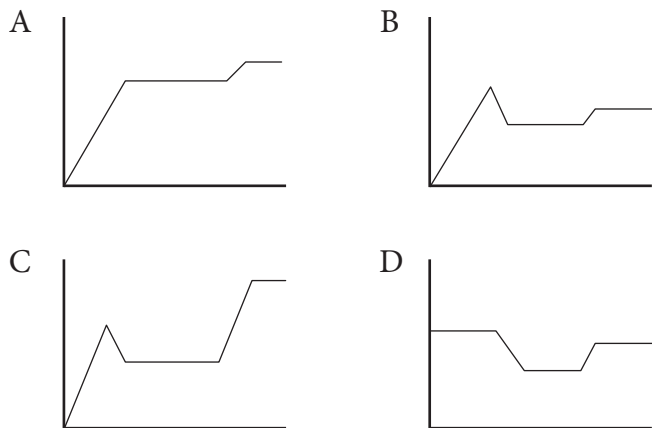


The most likely regions where he would accelerate are between points:

- A** (i) R and S, and (ii) T and U.
B (i) R and S, and (ii) S and T.
C (i) P and Q, and (ii) T and U.
D (i) S and T, and (ii) T and U.
- 15** When the velocity of a person or object decreases with time, we say that there is a 'negative acceleration' or 'deceleration'. The places in the diagram in question 14 where the cyclist is most likely to experience a deceleration are between the points:

- A** Q and R and also R and S.
B S and T and also T and U.
C R and S and also S and T.
D Q and R and also S and T.

- 16** A driver starts his car, accelerates to a speed of 60 kmph, notices a '40 kmph school zone', slows to the correct speed and then regains a speed of 50 kmph which he maintains. Which graph best shows these changes?



- A** A **B** B **C** C **D** D

- 17** When we use the term 'a force' we mean:

- A** something that always causes something else to move.
B an occurrence that always causes major damage.
C simply a 'push' or a 'pull' on something.
D a 'push' being applied to an object which moves as a result.

- 18** When a quantity such as a force is to be measured, scientists usually make up a unit with which it can be measured and name it after an appropriate scientist who did work on the subject. In the case of the quantity we call 'force', the unit is named after:

- A** Henri Ampere.
B Sir Isaac Newton.
C James Watt.
D James Joule.

- 19** A single unit of force is roughly equivalent to the force you would use when you pick up a couple of metallic table knives. This means that the unit is:

- A** a reasonably small unit.
B a suitable unit to which we can add prefixes such as 'kilo' when we wish to measure bigger forces.
C a suitable unit to which we can add prefixes such as 'milli' when we want to measure smaller forces.
D all of the above.

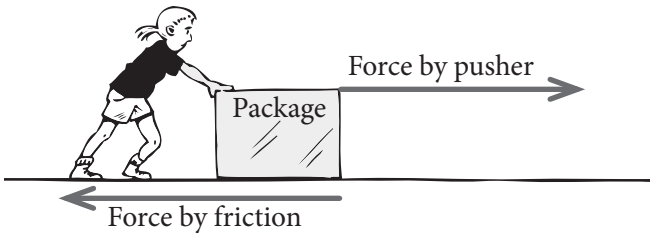
- 2|0** The table shows the approximate sizes of forces required to perform certain tasks.

Task	Approximate force needed
Lift a glass filled with water	5 N
Lift a small child	200 N
Accelerate a large jet to take-off	30 000 N

Considering these values, we could predict that the force required to push a stationary car to get it moving along a straight road would most likely be:

- ☐ A between 5 N and 200 N.
- ☐ B between 200 N and 30 000 N.
- ☐ C greater than 30 000 N.
- ☐ D about 250 N.

- 2|1** One force which opposes motion is the one called 'friction'.



A person pushing a package along a floor as shown in the diagram above would experience friction:

- ☐ A between the package and the floor and this would lessen the effect of the 'pusher's' force.
- ☐ B between his feet and the floor which would help him in trying to move the package.
- ☐ C which would always help the 'pusher' and make his job easier.
- ☐ D which would act in the ways described in A and B.

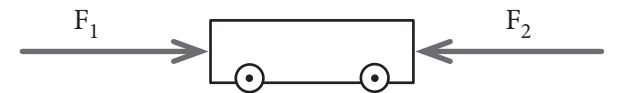
- 2|2** The force of friction is not an assisting one in which of the following situations?

- ☐ A A car driver applies the brakes.
- ☐ B A skier moves downhill, approaching a ski jump.
- ☐ C When we walk or run.
- ☐ D When an aeroplane accelerates rapidly down a runway.

- 2|3** When objects move through air, a force which opposes them is 'air resistance', which is a push by the air back on the objects as they push through the air. Sometimes this can be used to advantage. One such situation is:

- ☐ A when a fast bowler sends a straight ball to a batsman in cricket.
- ☐ B when a bullet is fired through the air.
- ☐ C the use of a parachute to slow a person's fall.
- ☐ D during the straight flight of a golf ball after it has been driven from the tee.

- 2|4** A force (F_1) pushes an object to the right and a second force (F_2) pushes the object to the left, at the same time. That is:



A true statement about the effect of these two forces is:

- ☐ A the object will only move if one force is greater than the other.
- ☐ B the object will move to the right.
- ☐ C the object will move to the left.
- ☐ D there cannot possibly be any movement of the object.

25 The 'mass' of an object is a measure of:

- A** how big the object is; that is, its size.
- B** the amount of matter that is 'packed into' the object.
- C** the volume of the object; that is, the space that it occupies.
- D** how close together or densely packed its particles are.

26 'Gravity' is the name given to the force of attraction which occurs between two objects, regardless of their sizes. It is always an attractive force. It is often hard to detect because:

- A** the objects concerned have relatively small masses and so the force is very small.
- B** the masses of the objects are so large that they tend to hide the effect of the force.
- C** it is restricted to certain situations only present on the Earth.
- D** the methods of detecting the force are extremely poor.

27 The size of the gravitational attractive force between two objects seems to:

- A** increase as the mass of either of the objects increases.
- B** increase as the two objects are moved further apart.
- C** decrease as the masses of the two objects are increased.
- D** decrease as the two objects are brought closer together.

28 When an object of mass 1.0 kg is near the Earth's surface, a force of approximately 10 N attracts it to the Earth. This means that:

- A** an object with a mass of 2.0 kg would be attracted to the Earth with a force of about 20 N.
- B** a 1.0 kg object also attracts the Earth with a force of about 10 N.
- C** a 2.0 kg object also attracts the Earth with a force of about 20 N.
- D** all of the above statements are true.

29 The force we call the 'weight' of an object is really a measure of the Earth's gravitational pull on the object, due to its mass. A girl with a mass of 50 kg would experience a force of approximately 500 N on her body. It follows that:

- A** every person would have a force of 500 N on his or her body.
- B** a boy with a mass of 30 kg would experience a force on his body greater than 500 N.
- C** a person of mass 100 kg would experience a force of about 1000 N on his or her body.
- D** a girl of mass 60 kg would experience a force of less than 500 N on her body.

30 The planet Jupiter is huge, compared to the Earth, and the gravitational attraction at its surface is about $2.6\times$ that of the Earth. This means that a person whose mass is 50 kg on Earth would experience, on Jupiter, a force on her body of approximately:

- A** $50 \times 10 / 2.6$ N.
- B** $50 \times 10 \times 2.6$ N.
- C** the same as that on Earth because of the great distance between Earth and Jupiter.
- D** 550 N this being a little more than that on Earth.

Questions 31–34 refer to the following information.

The table below shows comparisons between planets of the Solar System in regard to their (a) masses and (b) gravity at the surface of the planet. To simplify things, the Earth has been given a mass of 1 ‘unit’ and a gravitational pull at its surface, also of 1 (different) ‘unit’. The measurements of the other planets use these same units.

Planet	Mercury	Venus	Earth	Mars	Jupiter	Saturn	Uranus	Neptune
Mass	0.04	0.82	1.00	0.11	318.3	95.3	14.7	17.3
Gravity at surface	0.27	0.86	1.00	0.37	2.64	1.17	0.92	1.44

31 According to the table, the planet with a mass closest to that of the Earth is:

- ☐ A Venus.
- ☐ B Mars.
- ☐ C Uranus.
- ☐ D Neptune.

32 According to the table, the two planets with surface gravities most similar to that of the Earth are:

- ☐ A Venus and Neptune.
- ☐ B Saturn and Uranus.
- ☐ C Venus and Uranus.
- ☐ D Saturn and Venus.

33 The further away we are from the centre of a planet, the lower the gravitational force. The mass of Jupiter is huge compared to that of the Earth (318×!) and yet its surface gravity is only 2.64× that of the Earth. This is probably explained by the fact that:

- ☐ A Jupiter is so far away from the Earth that its gravity is lowered.
- ☐ B the surface temperature of Jupiter is much lower than that of the Earth.
- ☐ C Jupiter is much further from the Sun than the Earth.
- ☐ D the radius of the Earth is much less than that of Jupiter.

34 Which statement is true?

- ☐ A A brick would have a bigger mass on Jupiter because of the planet’s huge mass.
- ☐ B A brick would have the same mass on Earth as on Jupiter.
- ☐ C Jupiter’s greater gravity would cause a brick to have a greater mass than it would on Earth.
- ☐ D The lower surface temperature of Jupiter would cause a brick to have a lower mass than on Earth.

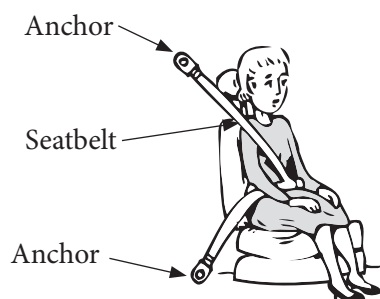
35 From the information in the preceding questions, we could deduce that the correct order of the heavenly objects, so far as their surface gravity is concerned, starting with the one with the lowest gravity and ending with the one with the greatest gravity, is:

- ☐ A Jupiter, the Earth and the Moon.
- ☐ B the Earth, the Moon and Jupiter.
- ☐ C the Moon, the Earth and Jupiter.
- ☐ D Jupiter, the Moon and the Earth.

36 When an object is given a small ‘push’ in space, it continues to move at a constant speed, virtually forever. This is:

- ☐ A because there is no force to change its motion and this is what Newton’s First Law of Motion states.
- ☐ B because the atmosphere in space is extremely thin.
- ☐ C because the original ‘push’ is always acting on the object.
- ☐ D expected because of the attraction of other planets.

- 37** A seatbelt in a motorcar is a safety device which prevents the person wearing it from being 'thrown forward' in the case of, for example, a head-on collision.



It is necessary because without the seatbelt, in such a collision:

- A** the seat applies a forward force which catapults the person forward.
- B** the person would continue to move forward at about the pre-collision speed of the car.
- C** the inertia of the person is lost.
- D** the person experiences an initial force backwards and so should be restrained to prevent injury.

- 38** The situation in question 37 is really a simple example of:

- A** Newton's Law of Universal Gravitation.
- B** Newton's First Law of Motion.
- C** Newton's Second Law of Motion.
- D** Newton's Third Law of Motion.

- 39** When a 'net' force is present, the object in question:

- A** will experience an acceleration.
- B** will only accelerate if there is no friction.
- C** usually undergoes a large acceleration.
- D** may accelerate if the slope is not too steep.

- 40** The situation in question 39 is really an example of:

- A** Newton's Law of Friction.
- B** Newton's First Law of Motion.
- C** Newton's Second Law of Motion.
- D** Newton's Third Law of Motion.

- 41** The term 'momentum' is often described as being the 'quantity of motion' which an object possesses. Alternatively, it can be considered as a measure of the 'crash effect' of the object. It is determined by the size of (a) the mass of the object and (b) the velocity of the object. A value for it is calculated by finding the product of the mass and the velocity. This means that:

- A** a stationary object has neither momentum nor inertia.
- B** a stationary object has inertia, but no momentum.
- C** a moving object has momentum, but no inertia.
- D** a moving object has inertia, but no momentum.

- 42** The commonly used unit of momentum is:

- A** newton m.
- B** kg kmph.
- C** kg m.
- D** kg mps.

- 43** If a 1 tonne car travels at 16.7 mps (≈ 60 kmph) then it would have a momentum of:

- A** 16.7×10^3 kg mps.
- B** 16.7×10^3 tonne mps.
- C** 1.67×10^2 kg mps.
- D** 60×10^3 kg mps.

44 The principle theme of Newton's Second Law of Motion links the 'net force' applied to an object, the 'mass' of the object and the 'acceleration' of the object.

It is expressed in the formula:

Force = mass $\times$ acceleration.

This means that:

- A** the acceleration of an object decreases as the net force applied to it increases.
- B** if the mass of an object is increased, then the acceleration that a specific net force would produce is also increased.
- C** as the net force applied to an object increases, so does the acceleration.
- D** both the mass and the acceleration increase as the net force on the object is increased.

45 As a moving object slows down or loses speed, it would be correct to say that it:

- A** loses momentum and gains a positive force.
- B** has a 'negative acceleration' and loses momentum.
- C** has a 'negative acceleration' and gains inertia.
- D** loses momentum, but gains inertia.

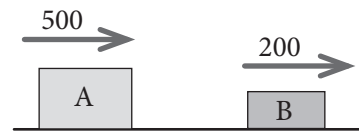
46 In order for a moving object to slow down:

- A** any force acting on the object would have to be removed.
- B** a frictional force is needed as it is the only one which will achieve this.
- C** some net force has to be applied to the object in the opposite direction to the way it is moving.
- D** the force due to air resistance has to be increased.

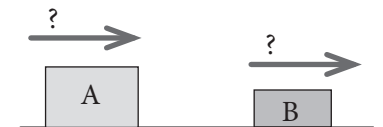
47 The 'Law of Conservation of Momentum' states that, in principle, when a collision occurs, the total amount of momentum which the colliding objects had before the collision is the same as the total amount of momentum which the objects have after the collision.

Consider the situation where a large object (A) with 500 units of momentum hits the back of a small object (B) travelling in the same direction with a lesser speed and with 200 units of momentum. That is:

Before collision



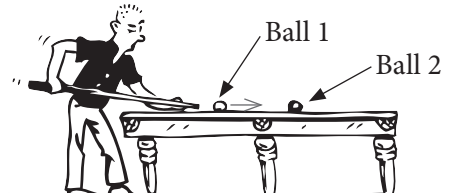
After collision



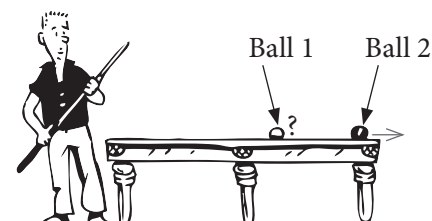
After the collision, the small object is boosted forward and now has more momentum. The result of this collision must be that:

- A** the combined momentum before the collision was 700 units so after the collision the combined momentum must be 700 units.
- B** if the momentum of B increased to 300 units, then the momentum of A would drop to 400 units.
- C** if the momentum of B increased to 400 units, then the momentum of A would drop to 300 units.
- D** all of the above statements would be true.

- 48** Consider the principle of ‘The Law of Conservation of Momentum’ applied in general terms to a pool game. In the following diagrams, a player strikes ball 1 with the cue and gives it 5 units of momentum. Ball 2, which has the same mass as ball 1, is stationary and is struck by ball 1.



CRACK!



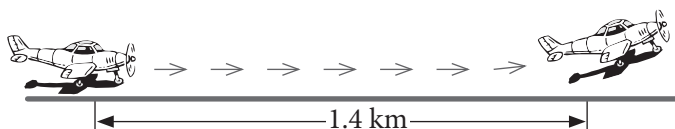
If, after the collision, ball 2 is sent forward in the same direction with 5 units of momentum then this would mean that ball 1:

- A** would rebound with 5 units of momentum in the opposite direction.
- B** would remain stationary.
- C** could keep rolling, but with a very small speed.
- D** probably rebounds in the opposite direction, but with a very small speed.

- 49** A small Cessna jet plane, ready for take-off, has a mass of 6700 kg. Each of its twin engines can supply a ‘thrust’ (forward force) of approximately 12.8×10^3 N. This means that when the plane is taking off and maximum thrust is used, the acceleration of the plane would be:

- A** $12.8 \times 10^3 \times 2 / 6700$ mpsps.
- B** $12.8 \times 10^3 / 6700$ mpsps.
- C** $6700 / 12.8 \times 10^3 \times 2$ mpsps.
- D** $6700 / 12.8 \times 10^3$ mpsps.

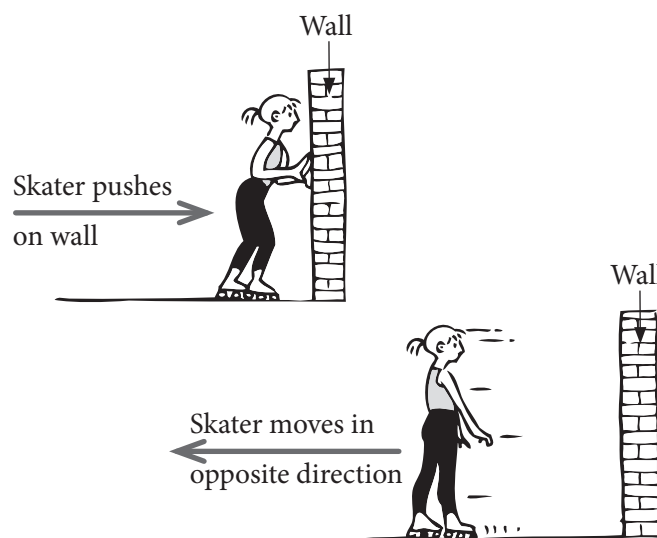
- 50** A large passenger jet with take-off mass 18 500 kg uses a runway of approximately 1.4 km to achieve its take-off speed.



Its average acceleration down the runway is about 2.2 mps. This means that the force required for the jet to take off is:

- A** in excess of 40 000 N.
- B** $18\,500 / 2.2$ N.
- C** approximately 4 000 N.
- D** a little less than 40 000 N.

- 51** A girl wearing roller skates and standing on a smooth surface pushes hard on a wall. As a result, she experiences a ‘reaction’ force and moves backwards. The wall, of course, doesn’t move. That is:



The reason for this is that:

- A** only the girl, having a smaller mass than the wall, experiences a force.
- B** the wall is connected to the Earth and so any movement caused by the girl’s push can’t be detected.
- C** the wall pushes harder on the girl than the girl pushes on the wall.
- D** with an object like an immovable wall, Newton’s Third Law of Motion does not apply.

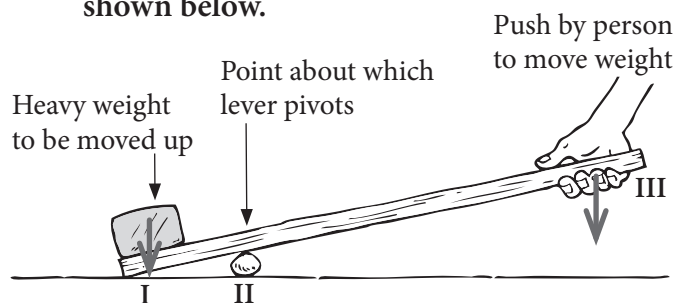
- 1** The Law of Conservation of Energy basically says that we can't obtain energy in one situation without losing it in another. In other words, we can't 'create' energy. This means that simple machines:

- A** always gain a little energy as they operate.
- B** basically change energy into a more usable form.
- C** make jobs easier by transforming energy from one form to another without any loss.
- D** make jobs easier because they give out more energy than they take in.

Levers

- 2** Perhaps the simplest machine is a lever. A lever usually takes the form of:
- A** a rigid bar (or equivalent) that pivots on a point and seems to make a task easier.
 - B** a long steel bar used for prising things.
 - C** a very short, hand-held device that assists people without needing a pivot.
 - D** all of the above.
- 3** Common devices which are simply levers include:
- A** a corkscrew, a wedge and handbrake handles on a bicycle.
 - B** an axe, a paddle and a handbrake handle on a car.
 - C** a ramp, a broom and an oar from a boat.
 - D** a broom, a fishing rod and handbrake handles on a bicycle.

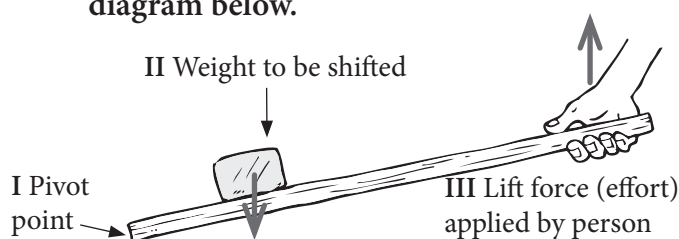
- 4** A simple situation involving a lever is shown below.



In general terms, the names that are used to describe the situations at points I, II and III in the diagram are, respectively:

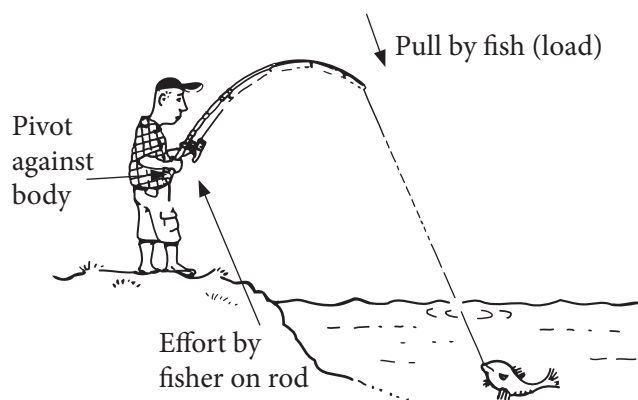
- A** load, effort and fulcrum.
 - B** load, fulcrum and effort.
 - C** effort, fulcrum and load.
 - D** effort, load and fulcrum.
- 5** In the situation in Question 4, the distances (or lengths) that we call the 'effort arm' and the 'load arm' are those, respectively, between:
- A** I and II, and II and III.
 - B** II and III, and I and III.
 - C** II and III, and I and II.
 - D** I and III, and I and II.
- 6** In a single lever system as in Question 4, it is true to say that, if the effort arm is of greater length than the load arm, then:
- A** the push force of the person will be less than the weight force to be moved.
 - B** the push force of the person may be less than the weight force to be moved, depending on the size of the weight force.
 - C** no matter what the weight force to be moved, the push force must always be greater.
 - D** the two forces will be of equal size and so 'balance' the weight force.

- 7** Another lever system is shown in the diagram below.



This arrangement of the pivot point, the weight to be shifted and the 'lift force' (effort) by the person:

- A** makes it harder for the person to shift the weight.
 - B** is not used in practice to shift a weight.
 - C** rarely makes it easier to shift a weight.
 - D** still makes it easier for the person to lift the weight.
- 8** While most levers try to make the job of moving a load easier (i.e. using an effort less than the load) one type of lever actually makes the job 'harder'. One example of such a lever is a fishing rod.

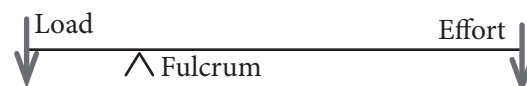


A lever such as the fishing rod is used:

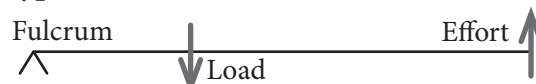
- A** when the load is large and has to be moved through only a small distance.
- B** when the load is small and needs to be moved through a greater distance than the effort of the fisherman.
- C** when the other types of levers cannot be used in the situation.
- D** when too much force is applied by the load for the other levers to cope.

- 9** The three types (or orders) of levers can be shown by the arrangement of the fulcrum, effort and load in each case. Diagrammatically, that is:

- Type 1 or 1st order lever



- Type 2 or 2nd order lever



- Type 3 or 3rd order lever



The reason Type 1 and Type 2 levers seem to make tasks easier is because:

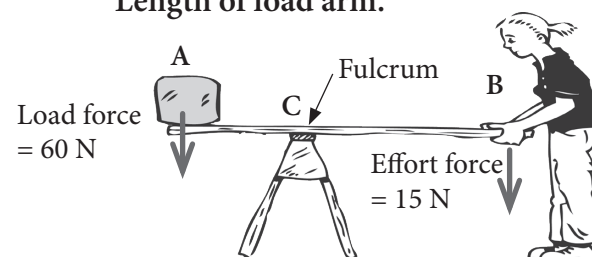
- A** the fulcrum is not necessarily at the end of the lever.
- B** of the direction of movement of load and effort.
- C** the effort arm in each case is longer than the load arm.
- D** the load arm in each case is longer than the effort arm.

- 10** The measure of a lever making a job usually somewhat easier is called the Mechanical Advantage (M.A.) of the lever. This is basically defined as:

$$\text{M.A.} = \text{Load force} / \text{Effort force.}$$

It is also measured by:

$$\text{M.A.} = \text{Length of effort arm} / \text{Length of load arm.}$$

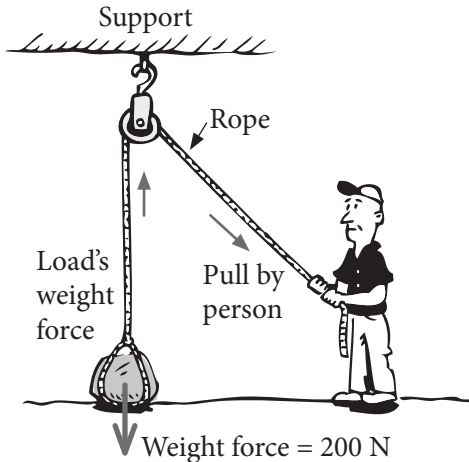


This means that, in the diagram above:

- A** the M.A. is 4.
- B** the M.A. is 1/4.
- C** the distance CB would have to be $2 \times$ the distance AC.
- D** the distance AC would have to be greater than the distance CB.

Pulleys

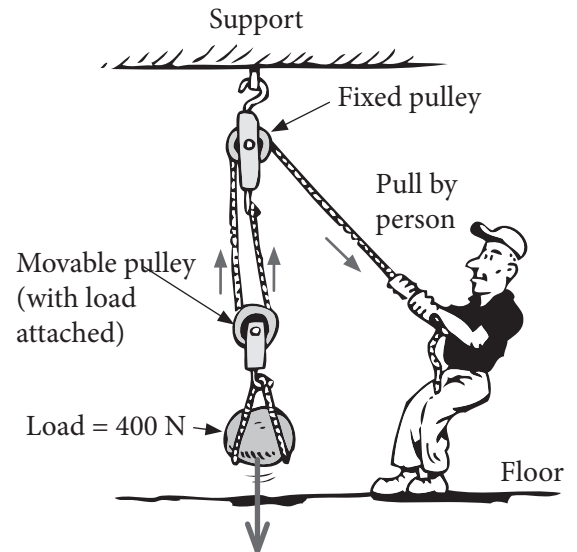
- 11** The machine called a 'single fixed pulley' is shown in the diagram where a load of 200 N is being lifted off the floor by a person pulling a rope.



In this situation, the advantage that the person gains is:

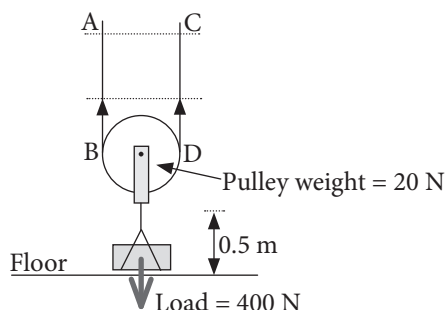
- A** because the pull required is less than 200 N.
 - E** simply a change in direction of the force needed to lift the load.
 - C** such that a pull of only 100 N is needed.
 - D** due to the pulley taking some of the load.
- 12** When the person does lift the load in question 11 off the floor, the pulley support would experience a force downwards of approximately:
- A** 200 N.
 - E** 600 N.
 - C** 100 N.
 - D** 400 N.
- 13** If in the situation shown in question 11, the load of 200 N is lifted 2 m above the floor, the amount of rope pulled through the pulley by the person would be:
- A** 4 m.
 - E** 1 m.
 - C** 2 m.
 - D** dependent on the size of the pulley.

Questions 14–16 refer to the following information. Another pulley system is the one called a 'single movable pulley'. It also involves the use of a 'single fixed pulley'. This arrangement is shown in the diagram, with a load of 400 N. The load is attached to the 'movable' pulley which has 2 ropes supporting it.



- 14** This arrangement means that the pull by the person:
- A** would be the same as with the 'single fixed pulley'.
 - E** is about 200 N, i.e. about half the load, since 2 ropes support the load.
 - C** is greater than 400 N.
 - D** is only $\frac{1}{3}$ of 400 N.
- 15** In this situation, the pull or effort by the person:
- A** is partially used in lifting not only the load, but the movable pulley too.
 - E** would move the load 2 m upward if 2 m of rope were pulled downward by the person.
 - C** could only move a maximum load of 400 N, no matter what the effort (pull) applied.
 - D** is restricted by the weight of the single fixed pulley.

- 16** The part of the diagram involving the load is shown in a separate diagram below. The goal is to lift the load, which is on the floor, by a distance of 0.5 m using the single movable pulley system.

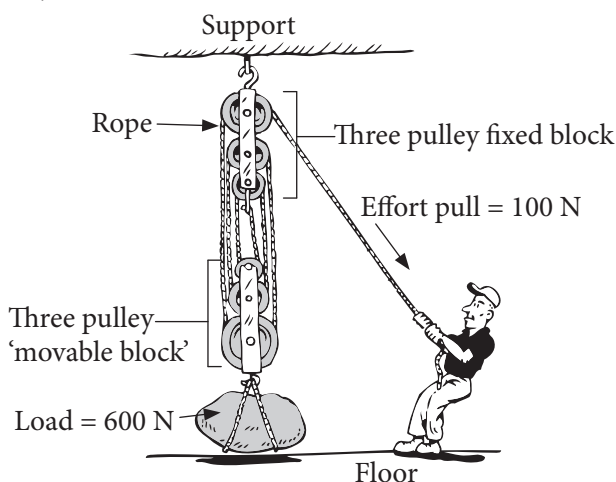


For this to be achieved:

- A** 0.5 m of rope has to be pulled down by the person.
- B** only the rope on side AB has to be shortened by 0.5 m.
- C** only the rope on side CD has to be shortened by 0.5 m.
- D** both the rope sections, i.e. sides AB and CD, have to be shortened by 0.5 m.

Questions 17–18 refer to the following information.

A single movable pulley makes the job of lifting heavy loads easier, because a smaller effort can be used. Therefore, this combination of pulleys can be expanded for even greater effect. A 'block' of fixed pulleys can be connected to a 'block' of movable pulleys to lift or move a very large load. In this example, there are 3 pulleys in each block, connected as shown in the diagram. (The pulleys are shown under one another, but are usually side by side). If we ignore the weight of the movable block, then the person exerting an effort of 100 N can just lift the load from the floor.



- 17** The effort required can best be explained by the fact that:

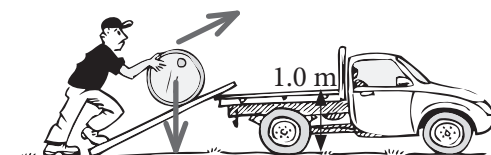
- A** the M.A. of the system is 6.
- B** each of the ropes supporting the movable block exerts an upward force of 100 N.
- C** both A and B are true.
- D** the M.A. of the system is 1/6.

- 18** In reality for this situation, if the movable block itself weighed 60 N the effort needed to lift the load + movable block would be:

- A** 160 N.
- B** 110 N.
- C** 90 N.
- D** still 100 N.

Inclined planes

- 19** The diagram shows a person rolling a heavy cylinder of weight 500 N up a ramp, onto the back of a truck.



This ramp: $Wt = 500 \text{ N}$

- A** would enable the person to move the cylinder up the ramp using a force less than 500 N.
- B** is not really what we would call a proper machine.
- C** still makes the effort by the person equal to 500 N.
- D** would have to be extremely long to give any mechanical advantage at all.

- 20** To obtain a M.A. > 1 for the situation in question 19, the:

- A** effort would have to be less than 250 N.
- B** effort would have to be more than 250 N but less than 500 N.
- C** length of the ramp would have to be at least 2 metres.
- D** ramp could be inclined to the road at any angle less than 90° .

21 Three examples of devices which give us some assistance because they are really versions of inclined planes are:

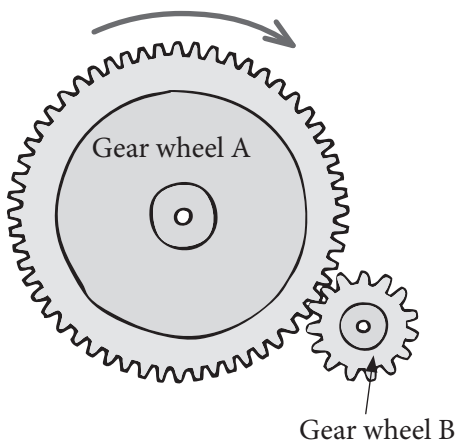
- ☐ A a corkscrew, a wedge and a knife.
- ☐ B a corkscrew, a hammer and a knife.
- ☐ C a wedge, a hammer and a knife.
- ☐ D a screwdriver, a wedge and an axe.

Gears

22 The purpose of using gears in any machine is to:

- ☐ A use less energy to shift a load.
- ☐ B always change the direction of an applied effort.
- ☐ C always increase the speed at which a load can be moved.
- ☐ D provide energy at a more suitable rate.

23 Gears often involve 'cogged wheels' (i.e. interlocking wheels) to transmit energy from one wheel to the other. Two such gear wheels are shown in the diagram. Gear wheel A has 60 'teeth' and gear wheel B has 15 'teeth'.



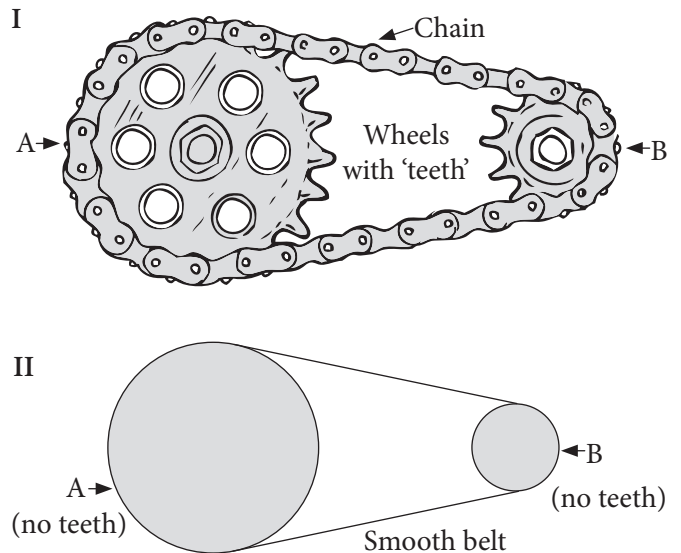
This means that, when gear wheel A turns clockwise one complete revolution, gear wheel B would turn:

- ☐ A clockwise through four revolutions.
- ☐ B clockwise through one revolution.
- ☐ C anticlockwise through four revolutions.
- ☐ D anticlockwise through one revolution.

24 In the diagram in question 23, it would be true to say that if an effort (E) were applied to rotate the cogs on wheel A, the load that could be moved by wheel B would be:

- ☐ A $2 \times E$.
- ☐ B $4 \times E$.
- ☐ C $\frac{1}{4} E$.
- ☐ D $\frac{1}{2} E$.

25 Sometimes gear wheels are not connected directly, but by a belt or chain which runs around them, as shown in the diagrams.



In situations like this, if gear wheel A rotates clockwise then gear wheel B will:

- ☐ A rotate anticlockwise at a faster rate in situation (i) but not in (ii).
- ☐ B rotate anticlockwise at a faster rate in situations (i) and (ii).
- ☐ C rotate clockwise at a slower rate in situations (i) and (ii).
- ☐ D rotate clockwise at a faster rate in situations (i) and (ii).

26 When a bicycle is being ridden on flat ground, the situation with a bicycle gearing system is most similar to that shown in question 25, diagram:

- ☐ A (i), where the pedals drive gear A.
- ☐ B (i), where the pedals drive gear B.
- ☐ C (ii), where the pedals drive gear A.
- ☐ D (ii), where the pedals drive gear B.

Efficiency

- 27** When the term 'efficiency of a machine' is used, it relates to the energy which goes into the machine compared to the energy which comes out of the machine. This is usually expressed as a percentage:

$$\text{Efficiency} = \frac{\text{Energy out}}{\text{Energy in}} \times \frac{100}{1}$$

If a machine has an efficiency of 30%, this would mean that:

- A** most of the energy put into the machine is not converted to a usable output.
- B** if 200 joules of energy went into the machine, then only 60 J would be available as usable energy output.
- C** the machine is very inefficient.
- D** all of the above statements are true.

- 28** The internal combustion engines which burn fuels such as petrol or diesel and are used to power most motor vehicles are generally very inefficient, usually well under 30% efficiency. The (i) energy conversion we are seeking and (ii) the main reason for the low efficiency, respectively, are:

- A** (i) chemical energy to kinetic and (ii) there is great heat loss in such engines.
- B** (i) chemical energy to kinetic and (ii) there is only a small amount of energy available from such fuels.
- C** (i) chemical energy to potential and (ii) there is great heat loss in such engines.
- D** (i) chemical energy to kinetic and (ii) the majority of energy from the fuel is converted to unwanted sound energy.

- 29** If a machine is described as 50% efficient then it would:

- A** prevent more than 50% of energy from being wasted.
- B** produce 150 joules of usable energy from an input of 300 joules of energy of a different type.
- C** lose half of the energy put into it because of conversion to unwanted forms of energy.
- D** produce the effects described in both B and C above.

- 30** For a particular machine, the energy input by way of the stored energy in a fuel and the energy outputs of different forms of energy are summarised in the table.

Energy input (J)	Energy output (J)
5000 as available energy stored in fuel	Kinetic: 1000 Heat: 3000 Sound: 600 Light: 400

The machine carrying out the conversion is:

- A** most efficient in producing kinetic energy.
- B** 20% efficient in producing kinetic energy.
- C** producing mainly kinetic, sound and light energy.
- D** only 10% efficient in producing light energy.

- 1** When we speak of the type of energy we call 'nuclear energy' as a source of usable energy, we are referring to a type which:
- A** can only be used to produce explosive forces.
 - B** comes from the nucleus of certain atoms.
 - C** is extracted only from the element uranium.
 - D** cannot be controlled successfully.
- 2** If an element is described as being 'radioactive', it means that the element:
- A** is automatically extremely dangerous.
 - B** will always be a good source of large amounts of nuclear energy.
 - C** spontaneously gives out certain radiations causing it to change to a different element(s).
 - D** will produce harmful radiations which maintain their intensities for millions of years.
- 3** The 'radiations' which come from radioactive elements are of three types: alpha (α), beta (β) and gamma (γ). Which of the following lists the relative properties of these radiations correctly?
- A** All types are highly penetrating and have similar masses.
 - B** β radiations are particles which are moderately penetrating, have a low mass and a + charge.
 - C** α radiations are particles which have a poor penetrating ability, negligible mass and a + charge.
 - D** γ radiations are highly penetrating, have no mass and no electrical charge.

- 4** The two types of nuclear reactions which release energy are called 'nuclear fission' and 'nuclear fusion'. In the following table, in which column are the features of these two types correctly shown?

Type	A	B	C	D
Fission	Involves 'splitting' of large nuclei of suitable elements	Involves 'splitting' of large nuclei of suitable elements	Involves forcing nuclei of small atoms together	Involves 'splitting' of large nuclei of suitable elements
Fusion	Involves forcing nuclei of small atoms together	Can involve splitting of large nuclei or forcing small nuclei together	Involves forcing nuclei of large atoms together	Involves splitting of small nuclei

- A** A **B** B **C** C **D** D

- 5** One major advantage of nuclear energy over conventional sources of energy (e.g. coal and oil) is that:
- A** it does not produce the 'greenhouse' gases that coal and oil do.
 - B** it is the safest form of energy to produce.
 - C** nuclear energy is much cheaper to produce than energy from conventional sources.
 - D** there are virtually no waste products for disposal.
- 6** One major disadvantage of nuclear energy is that:
- A** it will always be extremely expensive to produce.
 - B** the waste products from its production are usually radioactive and present a disposal problem.
 - C** the energy is difficult to transport large distances.
 - D** it can only be used in certain countries where the raw materials are found.

7 Nuclear energy is used to produce electricity in many countries of the world. The process involves a 'nuclear reactor' which provides energy to heat water and convert it into steam. The steam, under pressure, then drives a turbine generator. The 'fuel' used in such a reactor is usually uranium:

- A** because it is the only one from which we can extract such energy.
- B** since the nuclei of its atoms combine spontaneously, releasing energy.
- C** since its fission can be controlled and it is a naturally occurring element.
- D** as it produces almost no hazardous waste products.

8 A real problem with using a nuclear reactor to provide energy is that if something major goes wrong:

- A** enormous amounts of heat energy can be produced, killing all nearby living things.
- B** harmful, radioactive elements are produced and these may travel large distances in the atmosphere.
- C** any contamination of the land can take many years to correct.
- D** all of the above can be true.

9 Australia currently has:

- A** no nuclear reactors producing electricity but has big reserves of uranium ore.
- B** no nuclear reactors producing electricity because it has no reserves of uranium ore.
- C** a small number of nuclear reactors producing electricity as uranium reserves are very limited.
- D** only one nuclear reactor which produces electricity, at Lucas Heights in Sydney.

10 One of the great things that has resulted from nuclear energy science is the production of radioactive 'isotopes'. Radioactive isotopes of an element are:

- A** atoms of the element which always have equal numbers of protons and neutrons.
- B** different forms of the common atoms of the element which give out nuclear radiations.
- C** atoms of the element which are no different to the normal atoms but which possess more energy.
- D** able to be used only in medical situations.

11 'Radioactive isotopes' are produced from the nuclear reactor at Lucas Heights, in Sydney. These isotopes are used in medicine. Some isotopes can be injected into the body system and traced to detect regions of body abnormalities. This means that these particular isotopes:

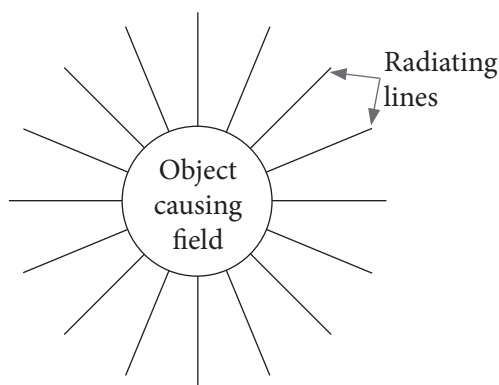
- A** are so mild that they cannot be detected.
- B** could not cause problems in the body no matter how strong their radiations were.
- C** do not harm body tissues in the concentration given and do not last long in the body.
- D** are always of elements which the body does not normally use.

12 When uranium atoms are 'split' to release nuclear energy, a particle is fired into the nucleus of the uranium atom to cause the split. The particle used for this purpose is:

- A** a neutron.
- B** a proton.
- C** an electron.
- D** an alpha particle.

- 1** In science, when we speak of a 'field' around an object, we mean:
- A** a two-dimensional area in which the object affects other objects.
 - B** a region surrounding the object in which the influence of the object can be detected.
 - C** a zone which, in some cases, becomes weaker with increasing distance from the object.
 - D** a zone which, in all cases, becomes stronger with increasing distances from the objects.

- 2** One way of representing a field in a two-dimensional diagram is to show lines radiating from the object causing the field, as shown in the diagram.



In this diagram, the lines indicate that the field:

- A** is stronger, closer to the object.
 - B** is weaker, closer to the object.
 - C** remains a constant, regardless of distance from the object.
 - D** is stronger on the right hand side of the object than on the left hand side.
- 3** The radiating lines in the diagram in question 2 are referred to as:
- A** lines of intensity.
 - B** lines of direction.
 - C** lines of magnitude.
 - D** lines of force.

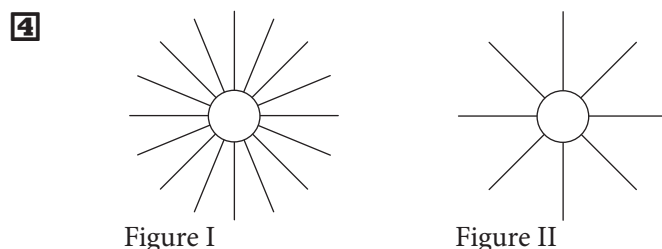


Figure I

Figure II

When two objects are shown with radiating lines as in Figures I and II, it means that both of the fields shown:

- A** are of the same type.
 - B** are of different types.
 - C** are of different strengths or intensities.
 - D** are of different types, but have the same strengths or intensities.
- 5** The following diagrams show different types of fields around particular objects causing the fields. The objects are not the same and the type of field is labelled in each case.

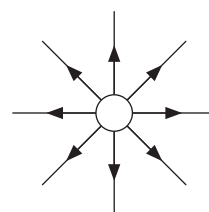


Diagram 1
Magnetic field

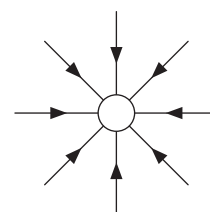


Diagram 2
Electric field

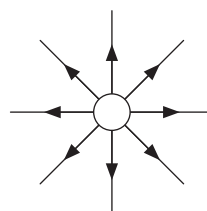


Diagram 3
Gravitational field

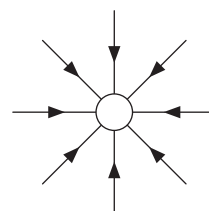


Diagram 4
Gravitational field

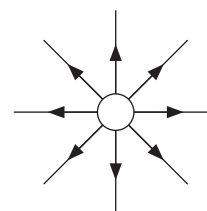
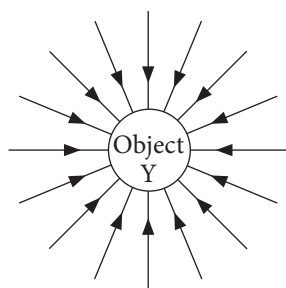


Diagram 5
Electric field

The diagram(s) which incorrectly show(s) the representation of a field:

- A** is Diagram 3 only.
- B** are Diagrams 1 and 3.
- C** is Diagram 5 only.
- D** are Diagrams 1 and 5.

6

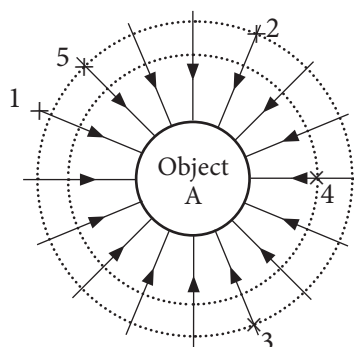


The radiating lines around the electrically charged object Y in the diagram above indicate that:

- A** a + charge in the field would experience a force acting away from object Y.
- B** a – charge in the field would experience a force acting away from object Y.
- C** a + charge in the field would experience a force towards object Y.
- D** both B and C are correct.

7

In the diagram below, 5 positions are labelled within the field created by object A.



The positions where the strength of the field is the same are:

- A** 1 and 3.
- B** 2, 5 and 4.
- C** 1, 2 and 3.
- D** 2, 3 and 5.

8

When an appropriate object is held in a position in the Earth's field, it is said to have 'potential energy'. The situation is similar:

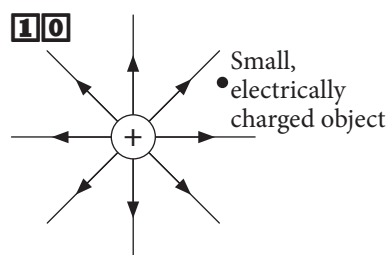
- A** if a charge is held in an electric field.
- B** when a magnetic substance, such as iron, is held in a magnetic field.
- C** if a charge is held in an electric field, but only if the charge is a positive one.
- D** in the situations described in A and B.

9

If a 1.0 kg mass is held in the Earth's gravitational field and then released and allowed to be attracted to the Earth, it will:

- A** lose its potential energy.
- B** have its potential energy converted to kinetic energy.
- C** experience a force which will make it accelerate towards the Earth.
- D** behave according to all of the above.

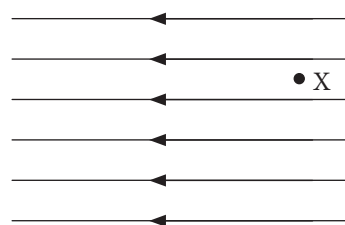
10



If a small, electrically charged object is allowed to move in the electric field shown in the diagram:

- A** it will move towards the positively charged object, regardless of its own charge.
- B** it will move either towards or away from the positively charged object, depending on its own charge.
- C** the field will make it move in a circular path around the positively charged object.
- D** it will move away from the positively charged object, regardless of its own charge.

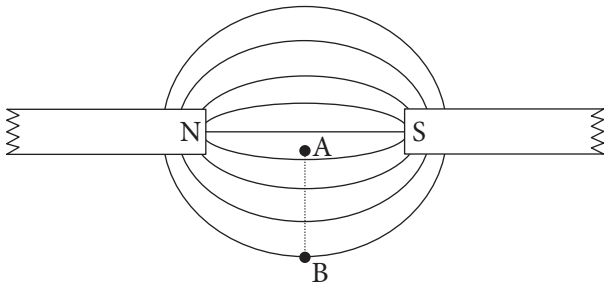
11



When a field is shown with parallel lines as in the diagram above, it means that:

- A** the field is a constant or uniform one, regardless of what type it is.
- B** the field becomes stronger from top to bottom in the diagram.
- C** the field can only be a magnetic field.
- D** the field can only be of the gravitational or electric types.

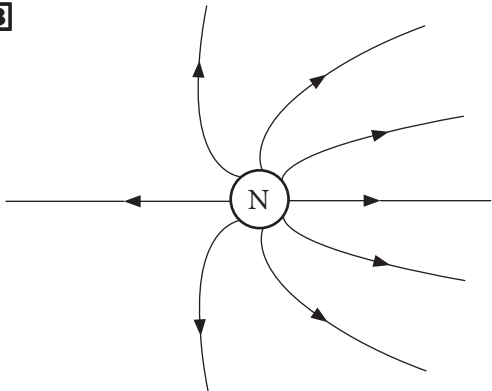
- 12** The north and south poles of two magnets are placed near one another to produce a magnetic field as shown in the diagram below.



If a sheet of paper was placed in the field, crossing it from A to B and at right angles to this page, then the effect on the shape of the field would be to:

- ☐ A alter it completely.
- ☐ B make the field move around the top of the paper at A.
- ☐ C change it in a negligible way since the magnetic field can easily penetrate paper.
- ☐ D push the field around point B and so make it stronger at that point.

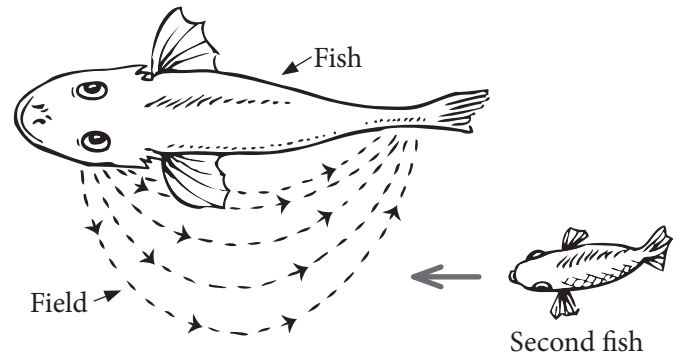
13



If a magnetic field around a north pole of a magnet appeared to be shaped as shown in the diagram above, it most likely means that the field was being influenced by:

- ☐ A another north pole to the right.
- ☐ B another north pole to the left.
- ☐ C either a north pole to the left or a north pole to the right.
- ☐ D either a south pole to the left or a south pole to the right.

- 14** A particular species of fish uses a 'field' around its body to detect changes in its environment, and hence potential predators or food sources. A portion of the field is shown below, viewed from above the fish.



If a second fish comes into the field, it causes changes to the field which the first fish can detect with special sensors on its body. From the diagram, it can be deduced that:

- ☐ A if the field around the fish is electric, then the fish's head behaves as a + charge and the tail behaves as a - charge.
- ☐ B if the field around the fish is electric, then the fish's tail behaves as a + charge and its head behaves as a - charge.
- ☐ C the field cannot be magnetic, since magnetic fields cannot penetrate water.
- ☐ D the field cannot possibly be either electric or magnetic.

- 15** The field around the fish in question 14 is probably caused by:

- ☐ A metal pollutants in its body.
- ☐ B external factors.
- ☐ C the Earth's magnetism.
- ☐ D special chemical reactions occurring inside the fish's body.

1 When a mirror is described as a 'curved' mirror, it usually means that:

- A** the surface of the mirror is part of a cylinder.
- B** the surface of the mirror is part of a sphere.
- C** the mirror is used to magnify objects.
- D** the surface of the mirror curves in different directions.

2 Curved mirrors are usually shown in cross-sectional views to indicate their shapes. Three such shapes are shown below.



Diagram I

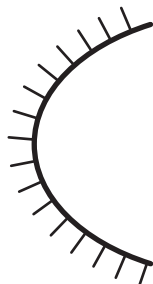


Diagram II



Diagram III

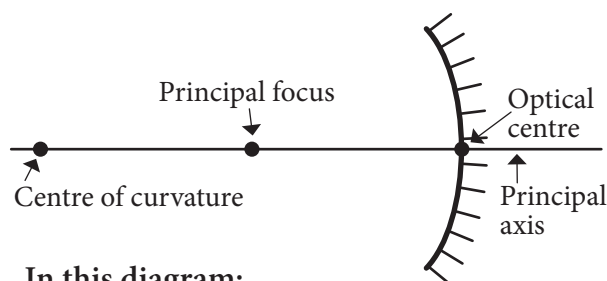
These diagrams indicate that the shapes of the mirrors are, respectively:

- A** concave, parabolic and concave.
- B** parabolic, concave and convex.
- C** concave, convex and parabolic.
- D** convex, parabolic and concave.

3 Curved mirrors are often used :

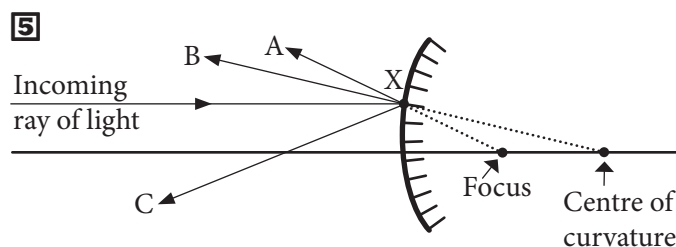
- A** internally in buses and coaches, as external rear vision mirrors on motor vehicles, in car headlights and torches.
- B** in buses and coaches, in car headlights, in torches and as bathroom wall mirrors.
- C** on bicycles as rear vision mirrors, as bathroom mirrors and in cameras.
- D** as bathroom mirrors, in lighting in the home and in cameras.

4 The diagram shows a curved mirror in cross-section with labels to various parts or important points.



In this diagram:

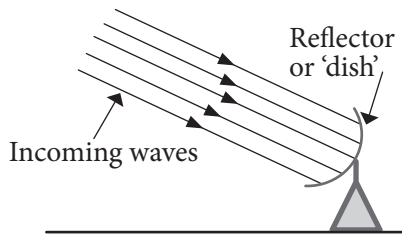
- A** the centre of curvature is twice the distance from the optical centre of the mirror than the principal focus.
- B** the principal axis of the mirror is incorrectly labelled.
- C** all rays of light striking the mirror would pass through the optical centre of the mirror.
- D** the labels 'centre of curvature' and 'principal focus' should be switched or reversed.



If an incoming light ray (or narrow beam) strikes a curved mirror as in the diagram above, it will be reflected according to the laws of reflection so that it will:

- A** pass back along the same path it travelled while approaching the mirror.
- B** travel along the line XB.
- C** travel along the line XA.
- D** travel along the line XC.

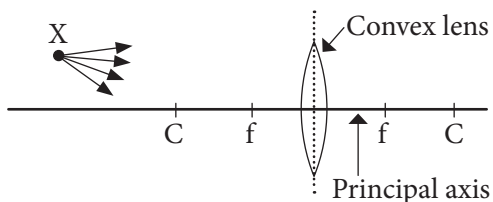
- 6** Curved reflectors are used to reflect waves other than light waves in many situations on a large scale. The set-up is usually as shown in the diagram.



Such reflectors are:

- A** usually convex and are used for microwaves.
 - B** always concave and are used exclusively for microwaves.
 - C** always concave and are used for radio waves and microwaves.
 - D** made of glass and focus incoming microwaves only.
- 7** Convex and concave lenses, respectively, are:
- A** thinner and thicker at the centre than at the edges.
 - B** both able to converge parallel rays of light.
 - C** able to converge and diverge parallel rays of light.
 - D** always made with cylindrical surfaces.

- 8** When incident rays of light leave an object at X in the diagram and pass through the convex lens shown, then the image of X formed would be:

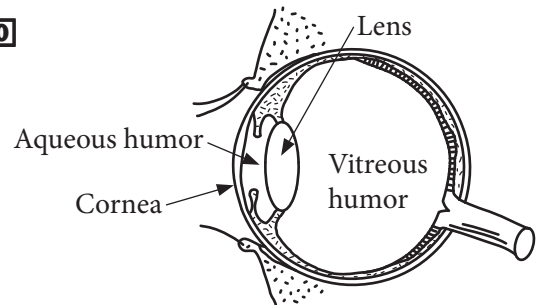


- A** enlarged, compared to the object at X.
- B** smaller than the object and inverted.
- C** further from the lens than the object X.
- D** on the opposite side of the lens to the object X and greatly magnified.

- 9** Lenses are used in:

- A** microscopes, telescopes and binoculars.
- B** telescopes, cameras and hand lenses.
- C** binoculars, cameras and microscopes.
- D** all of the above.

10



When light enters the eye, the major portion of the refraction needed to form an image on the retina occurs when the light passes from:

- A** the atmosphere into the cornea.
- B** the cornea into the aqueous humor.
- C** the aqueous humor into the lens.
- D** the lens into the vitreous humor.

- 11** The image of what we see is recorded by the human eye as:

- A** an inverted, real image which the brain interprets.
- B** a real, upright image which the brain interprets.
- C** an upright image on the retina of the eye.
- D** a virtual, inverted image on the retina of the eye.

- 12** The lens of the human eye is:

- A** able to change from convex to concave as required for focusing.
- B** a convex one which only causes minor changes in focus.
- C** is the major cause of refraction of light as it passes into the eye and to the retina.
- D** is the part of the eye responsible for us seeing in colour.

1 The production of sound energy:

- ☐ A is always accompanied by either light or heat energy.
- ☐ B is the result of some form of vibration.
- ☐ C results in very small pressure variations in the substance transmitting the sound.
- ☐ D involves both B and C above.

2 The table below shows the approximate speed of sound in some different substances, measured at the same temperature.

Substance	Iron	Granite	Air	Water
Speed of sound (metres per sec)	5130	6000	330	1400

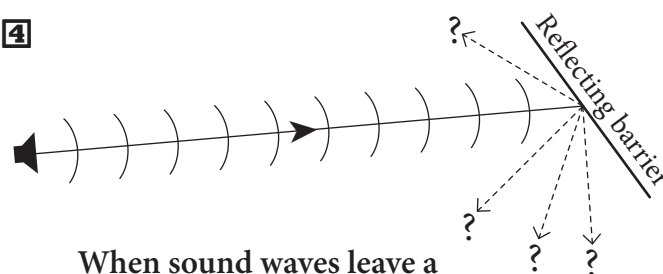
This table shows that the correct order of substances, in decreasing order of sound transmission speed within them, is:

- ☐ A air, water, iron, granite.
- ☐ B granite, iron, water, air.
- ☐ C iron, granite, water, air.
- ☐ D air, water, granite, iron.

3 Sound will not travel directly from person to person on the Moon because:

- ☐ A there is no atmosphere of any type to transmit sound.
- ☐ B the sensitivity of the human ear decreases on the Moon.
- ☐ C the air is too thin to transmit sound.
- ☐ D the Moon's temperature is so low that sound is not able to be transmitted.

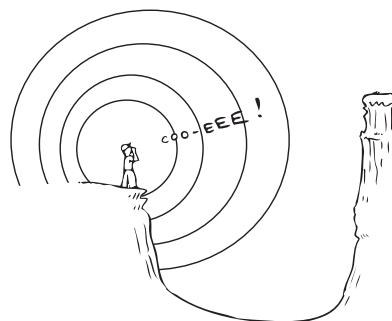
4



When sound waves leave a source and travel to a reflecting barrier, they are reflected:

- ☐ A according to the same rules that apply to light.
- ☐ B differently to the way in which light is reflected.
- ☐ C back along the incident path.
- ☐ D similarly to light, with the same speed.

5



When an 'echo' is heard from a person's voice, it means that the original sound:

- ☐ A has to be very close to what caused the echo.
- ☐ B has been amplified or increased in volume by a distant object.
- ☐ C has been reflected and the original sound is separated from the reflected sound.
- ☐ D has been reflected by some object in excess of 2.0 km from the source.

6

As sound travels from warm air to cold air, its speed decreases. This fact, by itself, would probably mean that:

- ☐ A warm air is more dense than cold air.
- ☐ B the cold air causes the wave length of the sound waves to increase.
- ☐ C the frequency of the waves would be much less in cold air.
- ☐ D cold air is more dense than warm air.

- 7** The information in the table below shows the speed of both light and sound waves in air and water. It also shows the densities of air and water.

Property	Air	Water
Density of substance	0.0013	1.00
Speed of light	3.0×10^8 mps	2.25×10^8 mps
Speed of sound	330 mps	1410 mps

The information in the table means that:

- A** light slows down as it passes from the less dense air to the more dense water.
B sound slows down as it passes from the less dense air to the more dense water.
C sound speeds up as it passes from the less dense air to the more dense water.
D both A and C above are true.
- 8** When a string or wire is 'stretched' (i.e. placed under tension) between two points as in Diagram 1, it can be pulled to one side and released.

Diagram 1

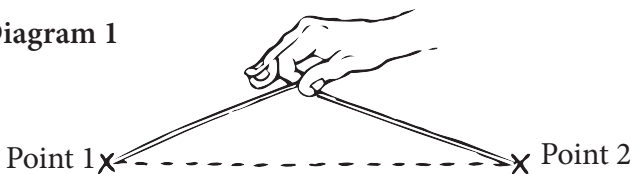
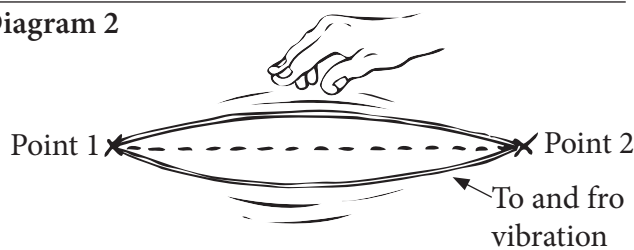


Diagram 2



This causes it to vibrate to and fro as shown in Diagram 2. This type of vibration produces a sound wave:

- A** if the frequency of the vibration is such that it can be detected by the human ear.
B regardless of its frequency of vibration.
C only if the amplitude of the vibration is extremely large.
D which is in no way similar to the sound wave produced by a vibrating air column.

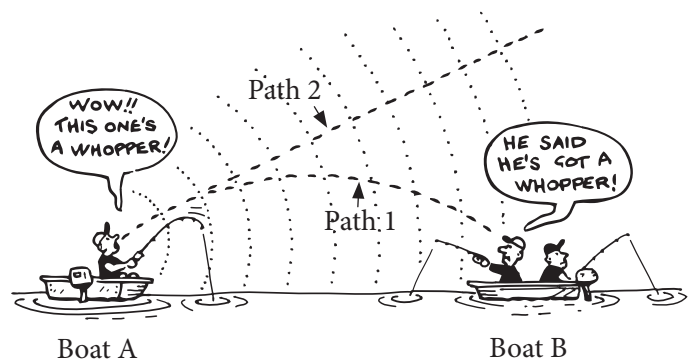
- 9** The wave equation $v = f\lambda$ applies equally to sound waves as it does to light waves. This means that if a sound wave travelling in air at 330 mps has a frequency of 512 Hz, then its wavelength would be, in metres:

- A** 330×512 . **B** $330 / 512$.
C $512 / 330$. **D** $330 + 512$.

- 10** Two musical instruments which create sound waves because strings vibrate and two which create sound waves because air columns vibrate are, respectively:

- A** double bass plus violin and clarinet plus piano.
B violin plus triangle and flute plus clarinet.
C harp plus double bass and trombone plus saxophone.
D guitar plus harp and piano plus trombone.

- 11** Sounds can often be heard quite clearly at night, particularly across large distances of water as shown in the diagram.



This is best explained by the fact that:

- A** at night, temperature changes cause more sound to travel from boat A to boat B via path 1 than path 2.
B sound waves are diffracted along path 1 at night time.
C more of the sound waves are reflected along path 2 than path 1.
D the cooler air above the water causes more sound to be absorbed along path 2 than in the daytime.

12 The approximate range of frequencies of sound which can be detected by the human ear (depending on age, health, etc.) is 20 Hz to 20×10^3 Hz. This means that, if the speed of sound in air is taken as 330 mps, then the range of wavelengths detected by the human ear would be:

- ☐ A 16.5 m to 16.5×10^2 m.
- ☐ B 1.65×10 m to 1.65×10^{-2} m.
- ☐ C 1.65 m to 1.65×10^{-2} m.
- ☐ D 1.65×10 m to 1.65×10^{-3} m.

13 The frequency of a sound interpreted by the human ear gives the sensation of 'pitch'. The pitch of a sound can be described as its 'highness' or 'lowness'. The greater the frequency, the higher the pitch. It follows that:

- ☐ A a bass drum gives out a sound which has a high pitch.
- ☐ B a piccolo gives out a sound which has a high pitch.
- ☐ C a piano gives out sounds that have a range of pitches.
- ☐ D both B and C above are correct.

14 The tiny vibrations received by the ear as 'sound waves' are magnified within the ear by some small bones which are called the:

- ☐ A magnifying bones.
- ☐ B enlargers.
- ☐ C ossicles.
- ☐ D stirrups.

15 The frequency range of vibrations detected by different animals is given below. The values are approximate only.

Animal	Frequency range of vibrations detected
Human	20 Hz – 20 000 Hz
Dog	50 Hz – 45 000 Hz
Mouse	1 000 Hz – 91 000 Hz
Dolphin	75 Hz – 150 000 Hz

From these values, it can be said that:

- ☐ A humans can hear low pitched sounds that a mouse cannot.
- ☐ B dolphins can detect an extremely wide range of frequencies, mostly in the higher range.
- ☐ C dogs can detect higher frequencies than humans can, but not lower ones.
- ☐ D all of the above are true.

16 A true statement about the 'loudness' of a sound is that:

- ☐ A it is determined by the wavelength of the sound wave and is commonly measured in units called 'decibels'.
- ☐ B both the wavelength and the amplitude of a vibration determine how loud a sound is.
- ☐ C the amplitude of the sound wave determines the loudness which is measured in units called 'bels' or smaller units called 'decibels'.
- ☐ D the amplitude of the sound wave determines the loudness which is measured in power units called 'watts'.

ANSWERS

Topic 1 – Waves

1 C **2** A **3** B **4** D **5** B
6 D **7** B **8** C **9** D **10** C
11 A **12** B **13** A **14** A **15** D
16 B **17** B **18** B **19** C **20** C
21 A **22** D **23** B **24** C **25** B
26 A **27** D **28** C **29** B **30** A
31 C **32** D **33** A **34** C **35** B
36 A **37** B **38** C **39** B **40** A
41 C **42** B **43** B **44** A **45** D
46 B **47** C **48** B **49** D **50** D
51 A **52** B **53** D **54** C

Topic 2 – Electricity

1 B **2** B **3** C **4** C **5** D
6 B **7** D **8** C **9** D **10** B
11 D **12** C **13** A **14** A **15** C
16 A **17** A **18** C **19** C **20** B
21 C **22** A **23** A **24** C **25** B
26 D **27** A **28** D **29** C **30** A
31 D **32** B **33** B **34** A **35** C
36 B **37** D **38** C **39** D **40** A
41 C **42** C **43** B **44** D **45** D
46 C **47** A **48** D **49** B **50** C

Topic 3 – Motion

1 B **2** C **3** A **4** D **5** A
6 B **7** B **8** C **9** D **10** A
11 B **12** D **13** C **14** A **15** D
16 B **17** C **18** B **19** D **20** B
21 D **22** B **23** C **24** A **25** B
26 A **27** A **28** D **29** C **30** B
31 A **32** C **33** D **34** B **35** C
36 A **37** B **38** B **39** A **40** C
41 B **42** D **43** A **44** C **45** B
46 C **47** D **48** B **49** A **50** A
51 B

Topic 4 – Machines

1 B **2** A **3** D **4** B **5** C
6 A **7** D **8** B **9** C **10** A
11 B **12** D **13** C **14** B **15** A
16 D **17** C **18** B **19** A **20** D
21 A **22** D **23** C **24** B **25** D
26 A **27** D **28** A **29** D **30** B

Topic 5 – Nuclear energy

1 B **2** C **3** D **4** A **5** A
6 B **7** C **8** D **9** A **10** B
11 C **12** A

Topic 6 – Fields

1 B **2** A **3** D **4** C **5** A
6 D **7** D **8** D **9** D **10** B
11 A **12** C **13** B **14** A **15** D

Topic 7 – Light

1 B **2** D **3** A **4** A **5** C
6 C **7** C **8** B **9** D **10** A
11 A **12** B

Topic 8 – Sound

1 D **2** B **3** A **4** A **5** C
6 D **7** D **8** A **9** B **10** C
11 A **12** B **13** D **14** C **15** D
16 C

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